MBFT 814 : Microbial Biotechnology 3(2-0-2)

M.Sc. Microbiology, Semester III · Complete detailed notes (full syllabus)

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Syllabus coverage

#	Chapter	Syllabus unit
1	Introduction and scope of microbial biotechnology	History and scope
2	History of microbial biotechnology	History and scope
3	Types of bioreactors	Bioreactors: packed bed, bubble column, airlift, fluidized bed, membrane / hollow fibre, photobioreactor
4	Types of fermentation	Batch, continuous, semisolid, solid state
5	Screening and strain improvement	Supporting topic (industrial strains)
6	Industrial alcohol (ethanol)	Industrial production of alcohol
7	Glycerol	Industrial production of glycerol
8	Solvents: acetone, butanol, isopropyl alcohol	Industrial production of solvents
9	Vinegar	Industrial production of vinegar
10	Organic acids: lactic, citric, gluconic	Industrial production of acids
11	Vitamins: riboflavin, B_{12}	Industrial production of vitamins
12	Antibiotics: penicillin, streptomycin, tetracycline, griseofulvin	Industrial production of antibiotics
13	Enzyme production and applications	Methods of enzyme production; therapeutic, analytical, industrial uses
14	Immobilization of enzymes	Enzyme immobilization
15	Single cell protein (SCP)	Microbes, substrates, nutritional value
16	Biotransformation	Biotransformation
17	Practicals	Fermenter, enzyme purification, lipase SmF and SSF, pH and temperature
18	Quick revision and question bank	All units

1. INTRODUCTION TO MICROBIAL BIOTECHNOLOGY

Definitions

- **Biotechnology**: the word was first used by the Hungarian engineer **Karl Ereky in 1919** for the production of products from raw materials with the help of living organisms.
- **OECD / Convention on Biological Diversity (1992)**: "Any technological application that uses biological systems, living organisms or derivatives thereof, to make or modify products or processes for specific use."
- **European Federation of Biotechnology**: "the integrated use of **biochemistry, microbiology and engineering sciences** in order to achieve technological (industrial) application of the capabilities of microorganisms, cultured tissue cells and parts thereof".
- **Microbial biotechnology**: the branch that uses **microorganisms** (bacteria, actinomycetes, fungi, yeasts, microalgae), their enzymes or their genes to make useful products (food, drugs, chemicals, fuels) or to give services (waste treatment, bioremediation) on a commercial scale.
- **Fermentation (industrial meaning)**: any process in which microorganisms are grown on a large scale to form a product, whether aerobic or anaerobic. (In the biochemical meaning, fermentation is anaerobic energy-yielding metabolism where organic compounds act as electron donor and acceptor.)

Why microorganisms are used

Property	Why it matters in industry
Small size, large surface : volume ratio	Very fast uptake of nutrients and release of products, so high metabolic rate
Rapid growth (generation time 20 min to a few hours)	Large biomass in hours or days instead of months (plants, animals)
Metabolic diversity	Can use sugars, starch, cellulose, hydrocarbons, methanol, CO_2; make acids, alcohols, enzymes, antibiotics, vitamins, polymers
Grow on cheap raw materials	Molasses, corn steep liquor, whey, bran, agro-industrial wastes lower the cost
Easy genetic manipulation	Mutation and recombinant DNA can raise yield many times
Controlled production	Independent of climate, season and land; uniform quality; easy scale-up in fermenters

Characteristics of an ideal industrial microorganism

- Pure culture, genetically **stable**, and easy to preserve for a long time.
- Gives a **high yield** of the required product in a **short time**.
- Grows on **cheap**, easily available medium; forms few unwanted by-products.
- Non-pathogenic, non-toxic, preferably **GRAS** (Generally Regarded As Safe).
- Produces spores or cells that are easy to use as **inoculum**.
- Product is easy to separate (preferably **extracellular**); resistant to contamination and phage.

Components (stages) of a fermentation process

Every industrial fermentation has six basic parts (Stanbury, Whitaker and Hall):

1. **Formulation of media** for the inoculum and production stages.
2. **Sterilization** of the medium, fermenter and air (medium: batch at 121°C, or continuous high-temperature short-time; air: filtration).
3. **Inoculum development**: stock culture → slant → shake flask → seed fermenter → production fermenter (usually 3–10% v/v inoculum).
4. **Production** in the fermenter under optimum pH, temperature, aeration and agitation.
5. **Downstream processing**: separation, extraction, purification and formulation of the product.
6. **Effluent treatment**: disposal of spent broth and biomass.

Steps 1–4 are called **upstream processing**; step 5 is **downstream processing**.

Fermentation media

Component	Examples	Role
Carbon source	Glucose, sucrose, **molasses**, starch, lactose, whey, vegetable oils, methanol	Energy and carbon skeleton
Nitrogen source	**Corn steep liquor (CSL)**, soybean meal, yeast extract, peptone, ammonium salts, urea	Proteins, nucleic acids
Minerals	P, K, Mg, S, trace Fe, Zn, Mn, Co, Cu	Enzyme cofactors, cell structure
Growth factors	Vitamins, amino acids (often supplied by CSL, yeast extract)	Needed by fastidious strains
Precursors	Phenylacetic acid (penicillin G), glycine (riboflavin), cobalt + DMB (vitamin B_{12})	Built directly into the product, raise yield
Inducers	Starch (amylase), oils (lipase), cellulose (cellulase)	Switch on enzyme genes
Buffers / antifoams	$CaCO_3$, phosphates / silicone oils, vegetable oils	Control pH / control foam

Types of microbial products

- **Microbial cells (biomass)**: baker's yeast, single cell protein (SCP), biofertilizers, probiotics, vaccines.
- **Microbial enzymes**: amylase, protease, lipase, cellulase, glucose isomerase.
- **Primary metabolites**: ethanol, citric acid, lactic acid, amino acids, nucleotides, vitamins.
- **Secondary metabolites**: antibiotics, pigments, toxins, alkaloids, gibberellins.
- **Recombinant products**: insulin, interferons, growth hormone, hepatitis B vaccine.
- **Biotransformation products**: steroid conversions (progesterone to 11α-hydroxyprogesterone by *Rhizopus*), vitamin C step by *Gluconobacter*.

Primary metabolites	Secondary metabolites
Made during active growth: **trophophase** (log phase)	Made after growth slows: **idiophase** (late log / stationary)
Essential for growth and reproduction	Not essential for growth; help survival (competition, defence)
Common to most organisms; produced in the same way	Restricted to a few species or strains
Production is growth-linked	Production is non-growth-linked; often repressed by glucose, phosphate
e.g. ethanol, citric acid, amino acids, nucleotides, vitamins	e.g. penicillin, streptomycin, tetracycline, pigments, mycotoxins

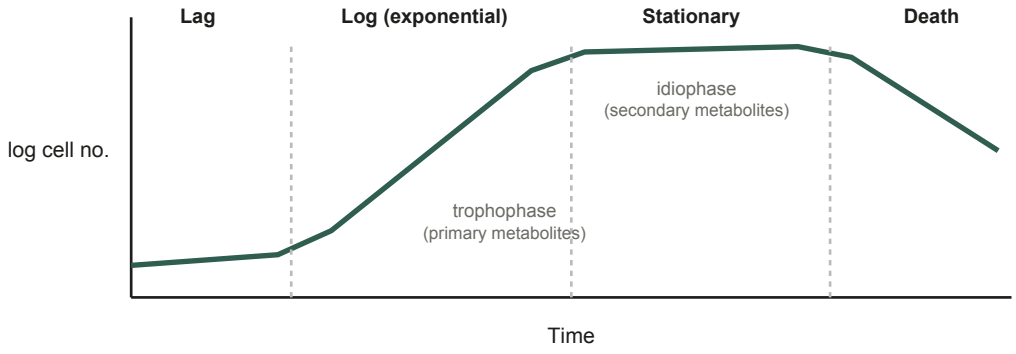


Fig. 1. Batch growth curve showing trophophase and idiophase

Scope and applications

Area	Examples
Food and beverages	Beer, wine, bread, cheese, curd, vinegar, soy sauce, SCP, amino acids (glutamic acid as MSG, lysine), flavours

Area	Examples
Health and pharmaceuticals	Antibiotics, vitamins, vaccines, insulin, interferons, monoclonal antibodies, steroids, therapeutic enzymes
Industrial chemicals	Ethanol, acetone, butanol, glycerol, citric, lactic, gluconic and itaconic acids, biopolymers (PHB, xanthan)
Enzymes	Detergent, starch, textile, leather, paper, dairy, juice industries
Agriculture	Biofertilizers (*Rhizobium*, *Azotobacter*, PSB), biopesticides (*Bacillus thuringiensis*, *Trichoderma*), silage, plant growth promoters
Environment	Sewage and effluent treatment, bioremediation of oil and pesticides, biogas, composting, bioleaching of copper and uranium
Energy	Bioethanol, biogas (methane), biodiesel, biohydrogen, microbial fuel cells

Exam point: For a long answer: definitions (Ereky 1919 + OECD) → why microbes → ideal industrial strain → six components of a fermentation process → primary vs secondary metabolites with growth curve → scope table.

2. HISTORY OF MICROBIAL BIOTECHNOLOGY

Eras of biotechnology

Era	Period	Features
Ancient biotechnology	Before 2000 BC (8000–4000 BC)	Domestication of plants and animals; fermented foods made by chance without knowing microbes
Classical biotechnology	2000 BC to 1900s AD	Deliberate fermentation for food and medicine; later scientific understanding of microbes (Leeuwenhoek, Pasteur, Koch)
Industrial / antibiotic era	1900–1953	Acetone–butanol, citric acid, penicillin, streptomycin; deep-tank fermentation; genetics
Molecular era	1953–1976	DNA structure, genetic code, restriction enzymes
Modern biotechnology	1977 onwards	Genetic engineering, recombinant proteins, PCR, genomics, metabolic engineering

Ancient and classical period

- **About 6000 BC**: Sumerians and Babylonians brewed **beer** with yeast.
- **About 4000 BC**: Egyptians made **leavened bread** using yeast; wine, vinegar and cheese were also made. Cheese began among nomadic tribes about 4000 years ago using **rennet** from calf stomach.
- Food was preserved by drying, salting, smoking and by fermentation (the acid formed stops spoilage).
- **1673–1676**: **Antonie van Leeuwenhoek**, with his single-lens microscope, first described bacteria and protozoa ("animalcules").
- **1701**: Giacomo Pylarini practised inoculation against smallpox; **1796**: **Edward Jenner** used cowpox to vaccinate against smallpox.
- **1809**: **Nicolas Appert** developed canning (heat sterilization of food in sealed containers).

The golden age of microbiology (1850–1900)

- **Louis Pasteur** (1822–1895), "father of industrial microbiology": showed in **1857** that lactic acid fermentation and later alcoholic fermentation are caused by **living microorganisms**; different microbes give different products; solved "diseases" of wine and beer; introduced **pasteurization**; described the **Pasteur effect** (oxygen inhibits fermentation); disproved spontaneous generation (swan-neck flask, 1861).
- **Robert Koch**: pure culture techniques on solid media and Koch's postulates (1876–1884), which made pure-culture industrial fermentation possible.
- **1877**: Wilhelm Kühne coined the word **"enzyme"** (Greek, "in yeast").
- **1897**: **Eduard Buchner** showed that a **cell-free yeast extract** ("zymase") could ferment sugar to alcohol. This proved fermentation is due to **enzymes**, not the "vital force" of living cells; Nobel Prize 1907.

Industrial fermentation era (1900–1950)

- **1914–1918**: **Chaim Weizmann** developed the **acetone–butanol–ethanol (ABE) fermentation** using *Clostridium acetobutylicum* on starch (maize). Acetone was needed to make **cordite** (explosive) during World War I. This was the first large-scale industrial fermentation needing strict aseptic conditions.
- **1917–1923**: J. N. Currie showed *Aspergillus niger* makes **citric acid** in surface culture; **Pfizer** started commercial production in 1923.
- **1919**: **Karl Ereky** coined "biotechnology".
- **1928**: **Alexander Fleming** noticed that *Penicillium notatum* killed *Staphylococcus* colonies and named the substance penicillin.
- **1938–1941**: **Howard Florey and Ernst Chain** (Oxford) purified penicillin and showed its clinical value (Nobel 1945 with Fleming).

- **1941–1945**: At the **Northern Regional Research Laboratory, Peoria (USA)**, penicillin production moved from surface flasks to **submerged deep-tank fermentation**: **corn steep liquor** raised yield, and the high-yielding strain *P. chrysogenum* **NRRL 1951** was found on a mouldy cantaloupe. This created modern fermentation technology (aeration, agitation, sterile air).
- **1944**: **Selman Waksman**, Albert Schatz and Elizabeth Bugie discovered **streptomycin** from *Streptomyces griseus*, the first drug effective against tuberculosis (Nobel 1952). Waksman coined the word "antibiotic".
- **1947–1950**: chloramphenicol (*S. venezuelae*), chlortetracycline (1948, Duggar, *S. aureofaciens*), and other antibiotics: "golden age of antibiotics".

1950–1976

- **1950s**: microbial **biotransformation of steroids** (e.g. 11α-hydroxylation of progesterone by *Rhizopus*, 1952) made cortisone cheap.
- **1953**: **Watson and Crick** proposed the double-helix structure of DNA.
- **1957**: **Kinoshita** (Japan) produced **glutamic acid** with *Corynebacterium glutamicum*: start of amino acid fermentation.
- **1960s–1970s**: **single cell protein** (SCP) from hydrocarbons and methanol (ICI "Pruteen" with *Methylophilus methylotrophus* in a large airlift fermenter); industrial enzymes such as glucose isomerase.
- **1969**: first industrial use of an **immobilized enzyme** (aminoacylase, Tanabe Seiyaku, Japan, I. Chibata).
- **1970**: discovery of **restriction enzymes** (Hamilton Smith); **1972**: Paul Berg made the first recombinant DNA molecule; **1973**: **Cohen and Boyer** cloned DNA in *E. coli* using a plasmid.

Modern biotechnology (1977 onwards)

Year	Event
1975	Köhler and Milstein: hybridoma technique for monoclonal antibodies
1976	Genentech founded (first genetic engineering company)
1977	Genentech produced **somatostatin** in *E. coli*: first human protein from a cloned gene
1978	Human **insulin** gene expressed in *E. coli*; rat insulin made by rDNA at Harvard
1982	**Humulin** (recombinant human insulin) approved: first rDNA drug
1983	Kary Mullis invented **PCR**
1986	First recombinant vaccine: **hepatitis B** surface antigen made in yeast; India set up the **Department of Biotechnology (DBT)**
1990–2003	Human Genome Project
1995	First complete bacterial genome: *Haemophilus influenzae*
1998	Remicade (monoclonal antibody for Crohn's disease) approved
2012 onward	CRISPR-Cas9 genome editing (Doudna and Charpentier); synthetic biology and metabolic engineering

Indian milestones: Hindustan Antibiotics Ltd., Pimpri (1954) for penicillin; IDPL (1961); DBT (1986); India is now a major producer of vaccines and generic antibiotics.

Exam point: Write history as eras with names and years. Must-know: Leeuwenhoek 1676, Pasteur 1857, Buchner 1897, Weizmann WWI, Ereky 1919, Fleming 1928, Florey–Chain and Peoria 1940s, Waksman 1944, Watson–Crick 1953, Cohen–Boyer 1973, somatostatin 1977, insulin 1982.

3. TYPES OF BIOREACTORS

Definition and functions

A **bioreactor** or **fermenter** is a vessel in which microorganisms, animal or plant cells, or enzymes are grown or used under **controlled, aseptic** conditions to carry out a biochemical process. Its function is to give the best environment (temperature, pH, dissolved oxygen, nutrients, mixing) so that the organism forms the maximum amount of product.

Requirements of an ideal bioreactor

- Able to run **aseptically** for long periods (sterilizable by steam in place).
- Good **aeration and agitation** to meet oxygen demand, without damaging cells (low shear).
- Low power consumption; **temperature control** (jacket or coils); **pH control**.
- Sampling, feeding and harvesting facilities; minimum evaporation loss.
- Foam control; easy cleaning and maintenance; made of **non-toxic, corrosion-resistant** material (stainless steel grade 316, glass for small units).
- Can be scaled up.

Parts of a stirred tank fermenter

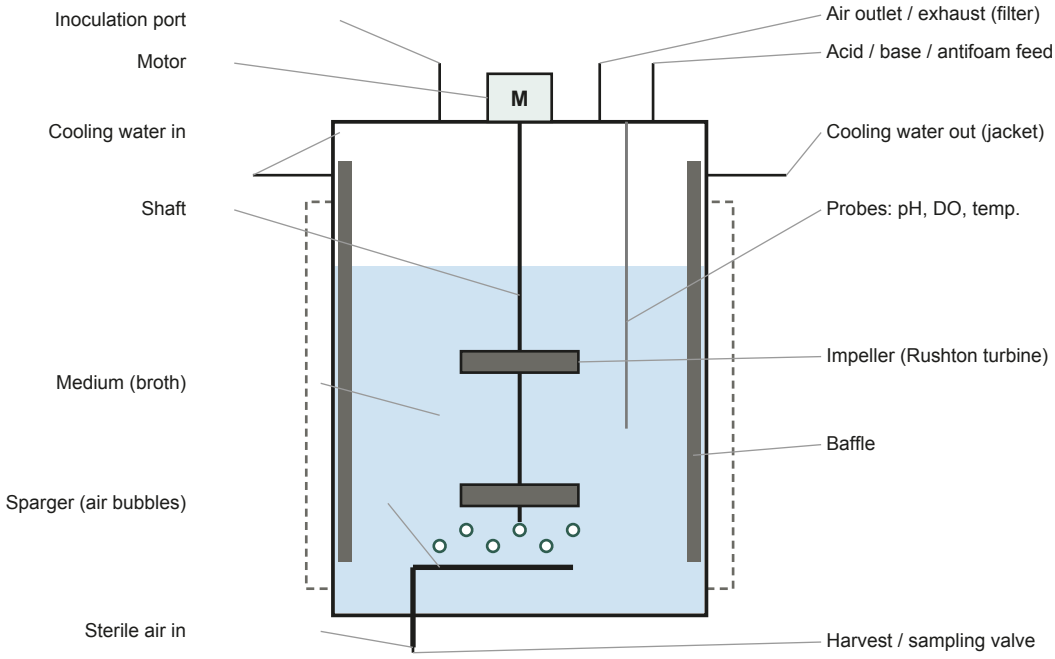


Fig. 2. Labelled diagram of a continuous stirred tank (stirred tank) fermenter

Part	Description and function
Vessel	Stainless steel cylinder with dished top and bottom; height : diameter about 2–3 : 1; working volume about 75–80% (headspace for foam)
Agitator (impeller)	Mounted on a shaft driven by a motor. **Rushton disc turbine** (flat blades on a disc) gives radial flow and breaks bubbles; marine propeller and pitched-blade turbines give axial flow. Mixes cells, nutrients and air and improves O_2 transfer
Sparger	Introduces sterile air at the bottom, below the impeller. Types: **porous** (sintered), **orifice** (perforated pipe or ring) and **nozzle** (open pipe)
Baffles	Usually **4 vertical metal strips**, width about 1/10 of vessel diameter, fixed to the wall; prevent **vortex** formation and improve mixing and aeration
Jacket / coils	Cooling or heating water circulates to keep temperature constant (removes metabolic and agitation heat)

Part	Description and function
Aeration system	Air compressor → sterile air filter → sparger; exhaust air through filter/condenser. Oxygen transfer is measured as $k_L a$
Foam control	Foam sensor with automatic addition of **antifoam** (silicone oils, vegetable oils) or mechanical foam breaker
Control probes	Temperature (thermocouple), **pH electrode** with acid/base pumps, **dissolved oxygen (DO) electrode**, pressure gauge, level sensor
Ports and valves	Inoculation, sampling, feeding (nutrients, precursor), and harvest valve; steam lines for sterilization in place

Types of bioreactors

(a) Continuous stirred tank reactor (CSTR)

- **Construction:** cylindrical vessel with motor-driven central shaft carrying one or more impellers, baffles, sparger, jacket (fermenter diagram above). Number of impellers depends on height.
- **Working:** air from the sparger is broken into small bubbles by the impeller and spread through the broth, giving a **uniform** environment.
- **Advantages:** efficient gas transfer, good mixing, flexible, easy control; the most widely used type.
- **Disadvantages:** high power use, high **shear** can damage filamentous fungi and animal cells, sealing of the shaft is a contamination risk.
- **Uses:** antibiotics, enzymes, amino acids, most aerobic fermentations.

(b) Bubble column bioreactor

- **Construction:** tall cylindrical column (height : diameter over 3 : 1) with a sparger at the base; **no mechanical agitator**; perforated plates may be placed inside to redistribute bubbles.
- **Working:** rising air bubbles both aerate and mix the medium.
- **Advantages:** simple, cheap, low shear, no moving parts, low contamination risk, low energy.
- **Disadvantages:** poor mixing at low gas flow, foaming, not good for viscous broth.
- **Uses:** baker's yeast, beer, vinegar, SCP, wastewater.

(c) Airlift bioreactor

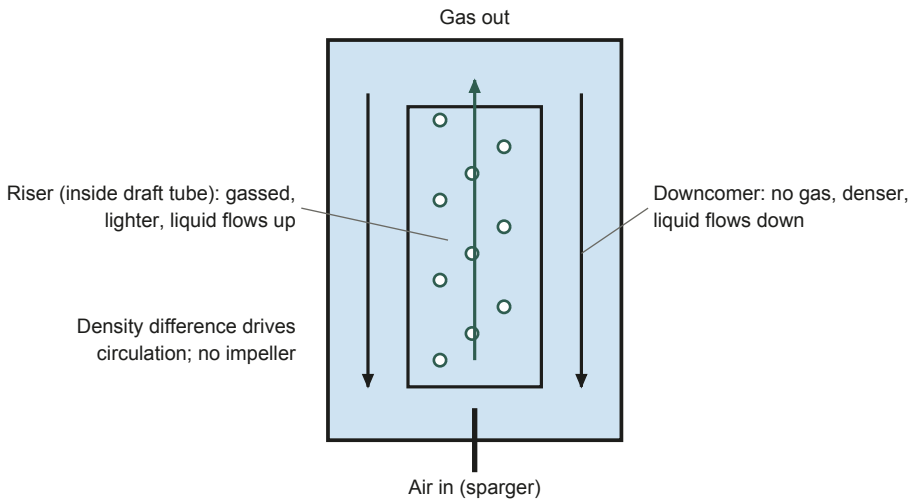


Fig. 3. Internal-loop airlift bioreactor (riser and downcomer)

- **Construction:** the vessel is divided into two zones by a **draft tube** or baffle: the **riser** (air is sparged into it) and the **downcomer** (no air). **Internal-loop** type has the draft tube inside; **external-loop** type has a separate outside tube connecting top and bottom.
- **Working:** the gassed liquid in the riser is less dense and rises; at the top gas escapes; the denser degassed liquid falls in the downcomer. This **density difference** causes circulation without any impeller.

- **Advantages:** low shear (good for plant and animal cells, fungi), low energy, good mixing and O_2 transfer, simple, easy to scale up, low contamination risk.
- **Disadvantages:** needs high air flow, less suitable for very viscous broths, foaming.
- **Uses:** SCP (ICI Pruteen plant, about 1500 m^3), plant and animal cell culture, monoclonal antibodies, wastewater treatment.

(d) Fluidized bed bioreactor

- **Construction:** similar to a bubble column but with a **narrow reaction column** and an **expanded top** section where fluid velocity falls, so particles fall back and are retained.
- **Working:** medium is pumped **upward** through a bed of small particles carrying immobilized cells or enzymes, fast enough to keep them **suspended (fluidized)**; liquid is often recycled.
- **Advantages:** good mixing and mass transfer, no clogging (unlike packed bed), suits immobilized cells, continuous operation.
- **Disadvantages:** hard to design and scale up, particles can be damaged.
- **Uses:** wastewater treatment (anaerobic), beer and vinegar with immobilized cells, immobilized enzyme reactions.

(e) Packed bed bioreactor

- **Construction:** a column packed with a **bed of solid particles** (porous or non-porous, rigid or compressible gels) with **immobilized** enzymes or cells on or inside them.
- **Working:** nutrient broth flows continuously through the bed (up or down); product leaves with the outflow. A **trickle bed** is a variant where liquid trickles down over the packing and air flows up.
- **Advantages:** high catalyst density, simple, continuous, high conversion, no need to separate catalyst from product.
- **Disadvantages:** poor mixing and O_2 supply, channelling, clogging, pH and heat gradients, compression of soft gels.
- **Uses:** immobilized glucose isomerase (HFCS), vinegar (trickling generator), effluent treatment.

(f) Membrane / hollow fibre bioreactor

- **Construction:** bundles of hollow semi-permeable fibres inside a shell. Cells are kept in the **shell (extracapillary) space**; medium flows through the fibres.
- **Working:** nutrients and oxygen diffuse out of the fibres to the cells; small products and wastes diffuse back into the flowing medium, while cells and large molecules are retained.
- **Advantages:** very high cell density, low shear, product separated continuously, cells protected.
- **Disadvantages:** membrane fouling, nutrient gradients, difficult scale-up, costly.
- **Uses:** animal cells, hybridomas (monoclonal antibodies), enzyme reactors, artificial organs.

(g) Photobioreactor

- **Definition:** a bioreactor with a **light source** for growing **photosynthetic** organisms (microalgae, cyanobacteria).
- **Open systems: raceway ponds** (shallow oval channels with a paddle wheel), cheap but easily contaminated, with poor control.
- **Closed systems: tubular** (transparent glass or plastic tubes), **flat-plate (panel)** and **column** reactors; good control of light, temperature, CO_2, and low contamination, but costly and can overheat.
- **Important factors:** light intensity and path (thin layers), CO_2 supply, O_2 removal (excess O_2 inhibits growth), temperature, mixing.
- **Uses:** *Spirulina*, *Chlorella* (SCP, health food), β-carotene (*Dunaliella*), astaxanthin (*Haematococcus*), algal biofuel.

Comparison of bioreactors

Bioreactor	Mixing by	Shear	Best for
Stirred tank (CSTR)	Impeller	High	Most bacterial and fungal fermentations
Bubble column	Rising air	Low	Yeast, simple aerobic processes
Airlift	Air + density difference (riser/downcomer)	Low	Shear-sensitive cells, SCP
Fluidized bed	Upward liquid flow	Medium	Immobilized cells, wastewater
Packed bed	Plug flow through bed	Low	Immobilized enzymes
Hollow fibre	Diffusion across membrane	Very low	Animal cells, antibodies
Photobioreactor	Air/pump + light	Low	Algae, cyanobacteria

Exam point: Always draw the stirred tank fermenter with at least 10 labels. For each other type write construction, working, advantages, disadvantages and uses. Airlift: remember riser, downcomer, draft tube, internal vs external loop.

4. TYPES OF FERMENTATION METHODS

Fermentations are classified by **mode of operation** (batch, fed-batch, continuous), by the **physical state of the medium** (submerged, surface, solid state, semi-solid), and by **oxygen requirement** (aerobic, anaerobic).

Batch fermentation (closed system)

- All nutrients are added at the start; the medium is sterilized and inoculated, and the fermentation runs to the end **without adding or removing** anything except air, acid/base for pH control and antifoam. At the end the whole broth is harvested.
- Cells pass through the **lag, log, stationary and death phases** (growth curve, Chapter 1). Nutrients fall and products and waste build up, so conditions keep changing.
- **Kinetics:** in the log phase $dX/dt = \mu X$, where μ is the specific growth rate. Doubling time $t_d = 0.693/\mu$. The **Monod equation** relates μ to substrate: $\mu = \mu_{max} \cdot S / (K_s + S)$.
- **Advantages:** simple, less risk of contamination and strain mutation, flexible (one plant for many products), suits **secondary metabolites** made in stationary phase.
- **Disadvantages:** "down-time" between batches for cleaning, sterilizing and filling; low overall productivity; batch-to-batch variation.
- **Examples:** beer, wine, many antibiotics, enzymes, vaccines.

Fed-batch fermentation

- A batch culture that is **fed** with fresh nutrient (or precursor) continuously or at intervals during the run, **without removing broth**; the volume increases, and everything is harvested at the end.
- **Why used:** keeps substrate at a low level and so avoids **substrate inhibition** (e.g. high glucose), **catabolite repression** (glucose repressing secondary metabolism) and the **Crabtree effect** in yeast (ethanol formed at high sugar); allows toxic precursors (phenylacetic acid) to be added slowly; extends the production phase; gives high cell density.
- **Types:** variable-volume (volume increases) and fixed-volume (substrate fed as concentrated solution); feeding can be constant, exponential, or controlled by DO or pH (feedback).
- **Examples: penicillin**, baker's yeast, many recombinant proteins, amino acids.

Continuous fermentation (open system)

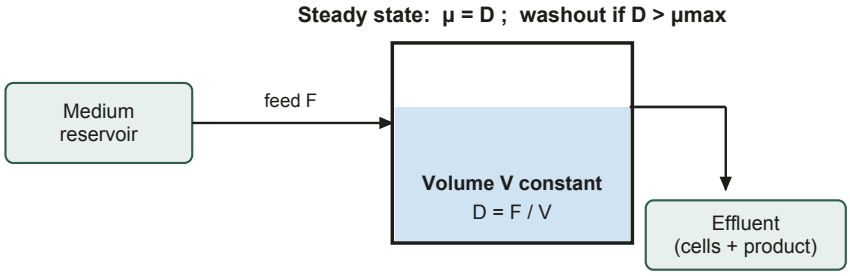


Fig. 4. Chemostat: fresh medium enters and broth leaves at the same rate

- Fresh medium is added **continuously** and culture broth is removed **at the same rate**, so the volume stays constant. After some time the culture reaches a **steady state** where cell, substrate and product concentrations stay constant and cells stay in the log phase.
- **Dilution rate** $D = F/V$ (F = flow rate, V = volume); units h^{-1}.
- Cell balance: $dX/dt = \mu X - DX$. At steady state $dX/dt = 0$, so **$\mu = D$**. The growth rate can therefore be set by the pump.
- If D is raised above μ_{max} (critical dilution rate), cells are removed faster than they grow and are lost: **washout**.

Type	Control	Notes
Chemostat	Constant flow of medium with one **growth-limiting nutrient**; growth rate set by D	Most common; stable at D below μ_{max}; used to study kinetics and physiology
Turbidostat	Cell density (turbidity) kept constant by a photocell that controls the feed pump	Works near μ_{max}; needs complex control
Multi-stage	Two or more vessels in series	First for growth, next for product formation (secondary metabolites)
With cell recycle	Cells separated (settling, membrane) and returned	Higher cell density and productivity; used in waste treatment and ethanol

- **Advantages:** high productivity, less down-time, uniform product, easier automation, cells kept in the most active state.
- **Disadvantages:** high risk of **contamination** over long runs, genetic instability and **back-mutation** of improved strains, not suitable for most secondary metabolites, wall growth, and difficult with filamentous organisms.
- **Examples:** SCP (Pruteen), ethanol, baker's yeast (some plants), vinegar, activated sludge process.

Feature	Batch	Fed-batch	Continuous
System	Closed	Semi-closed	Open
Volume	Constant	Increases	Constant
Nutrient addition	Only at start	During run	Continuous
Product removal	At end	At end	Continuous
Growth phase	All phases	Extended log / production	Steady log phase
Contamination risk	Low	Medium	High
Productivity	Low	High	Highest
Example	Beer, antibiotics	Penicillin, baker's yeast	SCP, ethanol

Submerged fermentation (SmF)

- Microorganisms grow **dispersed in a liquid medium** with aeration and agitation, in a stirred tank or airlift fermenter.
- **Advantages:** easy control of pH, temperature, O_2; uniform conditions; easy scale-up and automation; used for most industrial products.
- **Disadvantages:** high energy and water use, product is diluted, more effluent.
- **Examples:** antibiotics, most enzymes, citric acid (modern), vitamins, amino acids.

Surface (static) fermentation

- Organism grows as a **mat or pellicle on the surface** of a still liquid medium in shallow trays or pans; no agitation.
- **Examples:** early citric acid production by *A. niger* in aluminium trays, the Orleans method of vinegar, early penicillin in flasks.
- Now largely replaced by submerged fermentation because of low productivity and large space needs.

Solid state fermentation (SSF)

- Growth of microorganisms on **moist solid substrates** in the **absence or near absence of free-flowing water**. The solid gives both support and nutrients.
- **Substrates:** wheat bran, rice bran, rice straw, sugarcane bagasse, corn cobs, fruit pulp, oil cakes.
- **Organisms:** mainly **filamentous fungi** (*Aspergillus*, *Rhizopus*, *Trichoderma*, *Penicillium*), as they tolerate low water; some yeasts and bacteria.
- **Important factors:** moisture content (about 40–80%) and **water activity** (a_w), particle size, aeration, temperature (heat build-up is the main problem), pH, inoculum.

- **SSF bioreactors:** tray, packed bed (column), rotating drum, stirred and fluidized-bed reactors.
- **Examples: koji** (*Aspergillus oryzae* on rice/soybean for soy sauce and sake), tempeh (*Rhizopus*), mushroom cultivation, composting, enzymes (amylase, cellulase, pectinase, protease), gibberellic acid, fungal spores for biopesticides.

Semi-solid fermentation

- Intermediate between SmF and SSF: the substrate is a **thick slurry or paste** that has some free water but not enough to be a true liquid (e.g. bran with a higher water ratio).
- Combines some advantages of both: better mixing and heat removal than SSF, higher product concentration than SmF.
- Used for enzymes such as **lipase** and protease (syllabus practical: lipase by semi-solid fermentation).

Feature	Submerged (SmF)	Solid state (SSF)
Free water	Plenty	Little or none
Organisms	Bacteria, yeast, fungi	Mainly filamentous fungi
Aeration	Needs forced aeration and agitation	Air diffuses through spaces between particles
Control of pH, temperature	Easy and uniform	Difficult; gradients and heat build-up
Contamination	Bacterial contamination possible	Less (low water activity)
Product concentration	Low (dilute)	High (concentrated)
Energy, water, effluent	High	Low
Scale-up	Easy	Difficult
Examples	Penicillin, citric acid, enzymes	Koji, fungal enzymes, compost, mushrooms

Other classifications

- **Aerobic** (need O_2): antibiotics, citric acid, SCP, enzymes, vinegar. **Anaerobic**: ethanol, lactic acid, acetone–butanol, biogas.
- **Pure culture** vs **mixed culture** (curd, kefir, biogas, composting).

Exam point: Draw the growth curve and chemostat. Write μ = D at steady state and explain washout. Give the batch vs fed-batch vs continuous table and the SmF vs SSF table.

5. SCREENING AND STRAIN IMPROVEMENT

Screening

Screening is the use of highly selective procedures to **detect and isolate** microorganisms of industrial interest from a large microbial population. It separates the few useful isolates from thousands of useless ones in the least time and cost. Sources are soil (richest source, especially for actinomycetes and fungi), water, sewage, compost, decaying food, extreme environments, and culture collections (ATCC, MTCC Chandigarh, NCIM Pune, NRRL).

Primary screening (qualitative)

Detects **which** organisms make the desired product. It is quick, simple and done on a large number of isolates, usually on agar plates.

(a) Crowded plate technique (for antibiotic producers)

- Soil is diluted and plated so that plates are **crowded** with 300–400 colonies.
- Colonies producing antibiotics show a **clear zone of inhibition** around them where other colonies do not grow.
- These colonies are picked and purified. The technique is simple but only detects producers active against neighbouring colonies.

(b) Test organism / cross-streak and agar plug methods

- **Crowded plate with test organism**: after growth the plate is overlaid with soft agar containing a test bacterium (e.g. *Staphylococcus aureus*); zones show antibiotic activity against it.
- **Cross-streak**: isolate streaked in one line; test organisms streaked at right angles after growth; inhibition near the isolate shows its **spectrum**.
- **Giant colony technique**: a single large colony is grown and tested against many organisms.

(c) Auxanography (for growth-factor producers)

- Used to find organisms that make **vitamins or amino acids**.
- Plate seeded with a test organism that **needs** that growth factor; isolates are spotted on it. A **zone of growth** of the test organism around an isolate shows it produces the growth factor.

(d) Indicator dye / plate assay methods

Product	Plate method
Organic acids	pH indicator dye (bromocresol green, bromothymol blue) or $CaCO_3$ in medium: yellow zone or clear zone of dissolved $CaCO_3$
Amylase	Starch agar flooded with iodine: clear zone on blue background
Protease	Skim milk or casein agar: clear zone
Lipase	Tributyrin agar: clear zone; or Rhodamine B–olive oil plate (orange fluorescence)
Cellulase	Carboxymethyl cellulose (CMC) agar stained with Congo red: clear zone
Phosphate solubilizers	Pikovskaya agar: clear halo

(e) Enrichment culture technique

- A medium is made that favours only the desired type, e.g. only the target compound as sole carbon source (for degraders), or selective temperature/pH (for thermophiles, alkaliphiles). Repeated subculturing enriches the wanted organism.

Secondary screening (quantitative)

Done on the few organisms selected in primary screening, usually in **shake flasks** or small fermenters. It tells **how much** product is made and whether the organism is worth developing.

- Measures **yield**, productivity and fermentation time; compares isolates.
- **Identifies** the organism (to check pathogenicity and whether the product is new or already known).

- Finds whether the product is new, and its chemical and biological properties (spectrum, toxicity).
- Determines optimum **pH, temperature, aeration, medium** components and the need for special conditions.
- Shows whether the product is intracellular or extracellular and whether the organism is genetically stable.
- Checks if the product is destroyed by the organism itself, and economics of the process.

Strain improvement

Wild isolates usually give very low yields. **Strain improvement** is the deliberate genetic change of a microorganism to make it better for industry.

Aims

- Increase **yield** and productivity; shorten fermentation time.
- Use of **cheaper** substrates; reduce unwanted by-products and foaming.
- Improve genetic stability, tolerance to product, and ease of downstream processing (e.g. extracellular product, pellet morphology).

(A) Mutation and selection (classical method)

Cells or spores are exposed to a **mutagen** to give about 90–99% kill; survivors are screened for improved mutants. Repeated rounds give stepwise improvement.

Mutagen	Type	Action
UV light (260 nm)	Physical	Forms **thymine dimers**; errors during repair (SOS) cause mutations
X-rays, γ-rays	Physical (ionizing)	Break DNA strands, cause deletions and rearrangements
NTG (N-methyl-N′-nitro-N-nitrosoguanidine)	Chemical, alkylating	Methylates guanine → GC to AT transitions; most potent, acts at replication fork
EMS (ethyl methane sulphonate)	Chemical, alkylating	Ethylates guanine → GC to AT transitions
Nitrous acid	Chemical, deaminating	Deaminates adenine and cytosine → transitions
Base analogues (5-bromouracil, 2-aminopurine)	Chemical	Incorporated in DNA and mispair
Acridine dyes (acridine orange, proflavine)	Chemical, intercalating	Insert between bases → frameshift mutations

Selection of mutants

- **Random screening**: test thousands of survivors in flasks for higher yield (used for penicillin).
- **Auxotrophic mutants**: need a growth factor; found by **replica plating** and enriched by penicillin enrichment. Block a pathway so an intermediate accumulates (e.g. homoserine auxotroph of *C. glutamicum* overproduces **lysine**).
- **Regulatory (analogue-resistant) mutants**: grown on a toxic **analogue** of the end product; resistant mutants often have enzymes insensitive to **feedback inhibition** and overproduce (e.g. AEC-resistant lysine producers).
- **Catabolite-derepressed** and **constitutive** mutants for enzymes; **revertants** of non-producing mutants that sometimes produce more.

(B) Recombination

- **Protoplast fusion**: cell walls removed by lysozyme (bacteria) or lytic enzymes (fungi) in an osmotic stabilizer (sucrose, mannitol, KCl) → protoplasts of two strains mixed with **polyethylene glycol (PEG)** → fusion → regeneration of cell wall on osmotically stabilized medium → selection of recombinants. Used for *Streptomyces*, *Penicillium*, yeasts.
- **Parasexual cycle** (Pontecorvo) in fungi without a sexual stage: heterokaryon → diploid → mitotic recombination → haploidization; used in *Aspergillus niger* and *P. chrysogenum*.
- **Transformation, transduction and conjugation** in bacteria; sexual hybridization in yeasts.

(C) Recombinant DNA technology and modern methods

- **Gene cloning**: extra copies of key genes, strong promoters, removal of repressors; expression of foreign genes in *E. coli*, *Bacillus*, yeast (insulin, chymosin, interferon).
- **Site-directed mutagenesis / protein engineering**: change a specific amino acid, e.g. oxidation-resistant subtilisin for bleach detergents.
- **Metabolic engineering**: block competing pathways and remove feedback control, e.g. riboflavin-overproducing *B. subtilis*.
- **Directed evolution**, **genome shuffling**, and genome editing (CRISPR).

Classic example: penicillin strain improvement

Strain	How obtained	Approx. yield
P. notatum (Fleming)	Original isolate, surface culture	About 2–20 units/mL
P. chrysogenum NRRL 1951	Natural isolate from a mouldy cantaloupe (Peoria)	About 100 units/mL (submerged)
NRRL 1951 B25	Single-spore selection	About 250 units/mL
X-1612	**X-ray** mutant	About 500 units/mL
Wis Q-176	**UV** mutant (University of Wisconsin); non-pigmented later mutants	About 900 units/mL
Modern industrial strains	Repeated mutation (NTG, UV) and selection	Over 50,000 units/mL (tens of g/L)

Preservation of industrial strains

- **Subculturing** on slants and storage at 4°C (short term); **overlaying with sterile mineral oil**.
- Storage as dried spores in **soil or silica gel** (fungi, actinomycetes).
- **Lyophilization (freeze-drying)**: cells frozen and dried under vacuum, stored for years.
- **Cryopreservation** in **liquid nitrogen (−196°C)** or at −80°C with glycerol or DMSO as cryoprotectant (best for long-term stability).

Exam point: Screening: define, then primary (crowded plate, auxanography, indicator plates, enrichment) and secondary (list 6 points). Strain improvement: aims → mutagens table → mutant selection → protoplast fusion → rDNA → penicillin example.

6. INDUSTRIAL PRODUCTION OF ALCOHOL (ETHANOL)

Introduction

Ethyl alcohol (ethanol, C_2H_5OH) is the oldest and largest product of industrial fermentation. It is used as **fuel** (blended with petrol; India targets E20, i.e. 20% ethanol in petrol), as an industrial **solvent** and raw material (acetaldehyde, acetic acid, ethyl acetate), in **alcoholic beverages** and in pharmaceuticals (hand sanitizers, tinctures). Most industrial ethanol today is made by fermentation of sugars by yeast; synthetic ethanol from ethylene is small.

Microorganisms

- ***Saccharomyces cerevisiae*** (distiller's yeast): the main organism; ferments glucose, sucrose, maltose; tolerates 12–15% ethanol; GRAS.
- *S. uvarum* (*S. carlsbergensis*, bottom-fermenting lager yeast); *Schizosaccharomyces pombe* (osmotolerant).
- *Kluyveromyces marxianus* and *K. fragilis*: ferment **lactose** (whey).
- ***Zymomonas mobilis***: bacterium using the Entner–Doudoroff pathway; faster fermentation, higher yield, less biomass, but a narrow substrate range and poor tolerance to inhibitors.
- For lignocellulose: engineered *S. cerevisiae* and *E. coli* that also ferment pentoses (xylose); *Pichia stipitis*; thermophilic *Clostridium thermocellum*.

Biochemistry

Yeast converts glucose to pyruvate by **glycolysis (EMP pathway)**. Pyruvate is decarboxylated by **pyruvate decarboxylase** to acetaldehyde + CO_2, and acetaldehyde is reduced by **alcohol dehydrogenase** (using NADH) to ethanol. The NAD^+ formed keeps glycolysis running under anaerobic conditions.

$C_6H_{12}O_6 \rightarrow 2\ C_2H_5OH + 2\ CO_2$ (theoretical yield: 0.511 g ethanol per g glucose; about 90–95% of this is achieved in practice because some sugar goes to cells and by-products such as glycerol).

- **Pasteur effect**: in the presence of oxygen yeast respires and ferments less, so production is run **anaerobically** (a little air at the start helps yeast growth and sterol synthesis).
- **Crabtree effect**: at high sugar concentration *S. cerevisiae* ferments even in air.

Raw materials (substrates)

Type	Examples	Pre-treatment
Sugary	**Cane molasses** (main in India, 45–50% sugar), beet molasses, sugarcane juice, fruit juices, whey	Dilute to 14–18% sugar, adjust pH 4.5–5.0 with H_2SO_4, add N (ammonium sulphate / urea) and P
Starchy	Maize, wheat, rice, sorghum, cassava, potatoes (grain-based distilleries)	Milling → cooking (gelatinization) → **liquefaction by α-amylase** → **saccharification by glucoamylase** (or malt) → glucose
Cellulosic (2G ethanol)	Bagasse, rice straw, corn stover, wood	Pre-treatment (steam explosion, dilute acid, alkali) to open lignin → enzymatic hydrolysis by **cellulases** → fermentation of glucose and xylose

Production process (molasses-based)

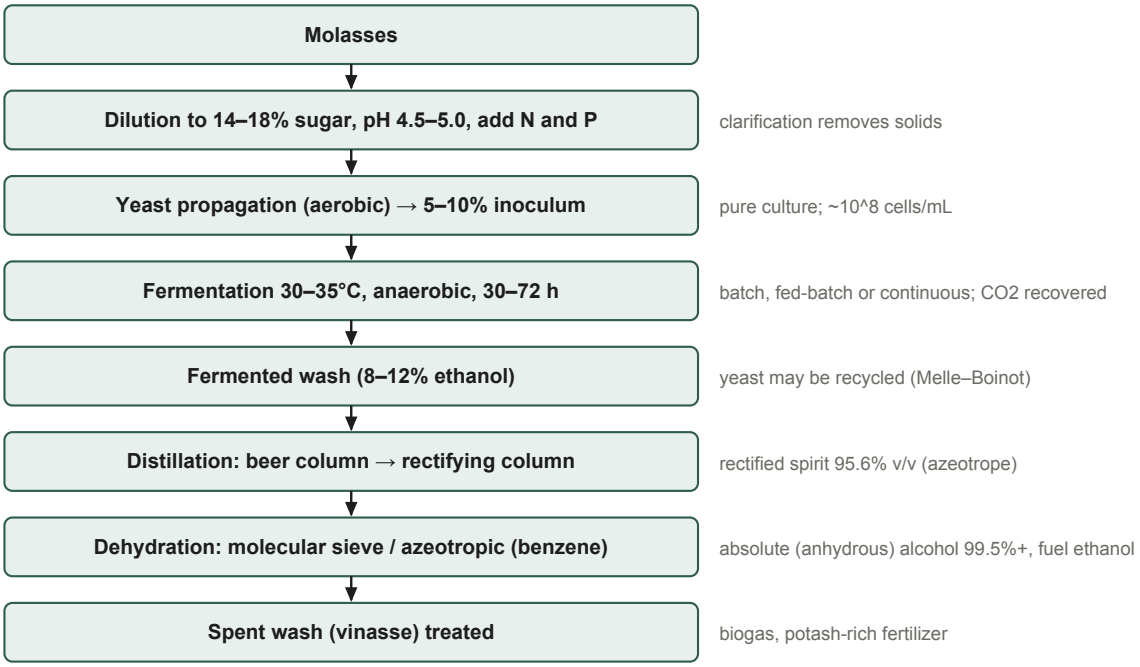


Fig. 5. Fermentation and recovery of industrial ethanol from molasses

- **Fermentation types:** batch (traditional), **Melle–Boinot process** (batch with **yeast recycling** by centrifugation: less sugar lost to yeast growth, shorter time), fed-batch, and **continuous** fermentation (multi-stage, often with cell recycle); immobilized yeast in packed/fluidized beds.
- **Conditions:** temperature 30–35°C (heat must be removed), pH 4.5–5.0 (limits bacterial contamination), anaerobic, 30–72 h; final ethanol 8–12% v/v.
- **Contaminants:** lactic acid bacteria (*Lactobacillus*), acetic acid bacteria, wild yeasts; controlled by low pH, hygiene and antibiotics/antiseptics.
- **By-products:** CO_2 (dry ice, beverages), **fusel oil** (higher alcohols: isoamyl, isobutyl, propyl alcohols), aldehydes, glycerol, yeast biomass (feed), spent wash (high BOD; biomethanation, composting, potash recovery).

Recovery (distillation)

- Fermented "wash" (beer) is heated in the **analyser (beer/stripping) column**, where vapours with 40–50% ethanol are separated from the spent wash.
- The vapour is concentrated in the **rectifying column** to **95.6% v/v (rectified spirit)**, which is an **azeotrope** with water and cannot be concentrated further by simple distillation. Fusel oil is drawn off from the middle trays; aldehydes (heads) come first.
- **Absolute alcohol** (99.5–99.9%) for fuel blending is made by **azeotropic distillation** (with benzene or cyclohexane) or, now mostly, **molecular sieve dehydration** (zeolites adsorb water).
- **Denatured alcohol**: made unfit for drinking by adding methanol, pyridine or dyes (industrial use, no excise duty).

Alcoholic beverages (brief)

Beverage	Raw material / organism	Key points
Beer	Barley malt, hops, *S. cerevisiae* (ale, top) / *S. uvarum* (lager, bottom)	Malting → mashing → boiling with hops → fermentation → lagering; 3–6% ethanol
Wine	Grape juice (must), *S. cerevisiae* var. *ellipsoideus*	Crushing, SO_2 added, fermentation 10–14%, malolactic fermentation (*Oenococcus*), ageing
Distilled spirits	Whisky (malted grain), rum (molasses), brandy (wine), vodka (grain/potato)	Fermentation then distillation (40–50% ethanol), ageing in oak
Indian products	Toddy (palm sap, yeasts + *Zymomonas*), handia (rice)	Traditional mixed-culture fermentations

Exam point: Write: definition and uses → organisms → biochemical pathway with equation and theoretical yield → substrates and pre-treatment → process flow chart → conditions and Melle–Boinot → distillation (rectified spirit 95.6%, azeotrope, absolute alcohol) → by-products.

7. INDUSTRIAL PRODUCTION OF GLYCEROL

Introduction

Glycerol (glycerine, propane-1,2,3-triol) is a sweet, viscous, non-toxic polyol used in pharmaceuticals (syrups, ointments, suppositories), cosmetics (moisturizer in creams and toothpaste), food (humectant, sweetener), tobacco, explosives (**nitroglycerine**, dynamite) and plastics. Most glycerol today is a by-product of soap making and of **biodiesel** production, but microbial glycerol was very important during **World War I**, when Germany (Connstein and Lüdecke, 1915) made about 1,000 tonnes per month by fermentation for explosives.

Principle: Neuberg's "steered" fermentations

In normal alcoholic fermentation only a small amount of glycerol (about 3%) is formed. **Carl Neuberg** showed that if acetaldehyde is removed or blocked, yeast must reoxidize its NADH by another route: it reduces **dihydroxyacetone phosphate (DHAP)** to glycerol-3-phosphate (glycerol-3-phosphate dehydrogenase), which is dephosphorylated to glycerol. So fermentation is "steered" from ethanol to glycerol.

Neuberg form	Condition	Result
First form	Normal acid pH, no additives	Glucose → 2 ethanol + 2 CO_2 (little glycerol)
Second form (sulphite process)	Add **sodium bisulphite / sulphite** (about 3–4%) at pH about 7	Sulphite binds **acetaldehyde** as an addition compound, so it cannot be reduced to ethanol; NADH is used to make glycerol: Glucose → glycerol + acetaldehyde-bisulphite + CO_2
Third form (alkaline process)	Alkaline pH 7–8 (Na_2CO_3 / $NaHCO_3$)	Acetaldehyde undergoes **dismutation (Cannizzaro)** to ethanol + acetic acid: 2 glucose → 2 glycerol + ethanol + acetic acid + 2 CO_2

Production (sulphite process)

- **Organism:** *Saccharomyces cerevisiae* strains tolerant to sulphite.
- **Medium:** molasses or sugar (about 10–20%), N and P salts, **sodium sulphite/bisulphite** added gradually (high sulphite is toxic).
- **Conditions:** about 30°C, pH about 7, anaerobic, 2–3 days. Yield about 20–25% of sugar as glycerol (theoretical maximum higher).
- **Recovery:** glycerol is non-volatile and hard to separate: remove cells, remove sulphite (as calcium sulphite with lime), evaporate, then **vacuum distillation** or steam distillation; ion exchange for purity. Recovery is the costly step.

Osmotolerant yeasts and algae (modern routes)

- **Osmotolerant yeasts** such as *Zygosaccharomyces rouxii*, *Debaryomyces hansenii*, *Candida glycerinogenes* and *Pichia farinosa* accumulate glycerol as a **compatible solute** when grown at high sugar or salt concentrations; aerobic processes give high yields without sulphite.
- The halotolerant alga ***Dunaliella*** accumulates very high intracellular glycerol in salt ponds.
- Genetically engineered *S. cerevisiae* (over-expressing GPD1 and deleting alcohol dehydrogenase) and *E. coli* produce glycerol.
- Glycerol from biodiesel is now so cheap that it is itself used as a **substrate** (e.g. for 1,3-propanediol by *Clostridium butyricum*, *Klebsiella*).

Exam point: Key idea: block acetaldehyde (sulphite) or remove it by dismutation (alkali) so NADH reduces DHAP to glycerol. Draw the branch point: DHAP → glycerol-3-P → glycerol vs acetaldehyde → ethanol.

8. INDUSTRIAL PRODUCTION OF SOLVENTS: ACETONE, BUTANOL, ISOPROPYL ALCOHOL

Introduction

Acetone (CH_3COCH_3), **n-butanol** (C_4H_9OH) and **isopropyl alcohol** (2-propanol) are important industrial solvents. Butanol is used in lacquers, paints, plasticizers, and is now studied as **biobutanol**, a better fuel than ethanol (higher energy, less water-absorbing, can be blended at higher levels). Acetone is used for explosives (cordite), plastics and as a solvent. The **acetone–butanol–ethanol (ABE) fermentation** was the second largest industrial fermentation after ethanol in the first half of the 20th century, until cheap petrochemicals replaced it in the 1950s; South Africa ran it till 1982, and China revived it in the 2000s.

History: the Weizmann process

Chaim Weizmann (Manchester, 1912–1915) isolated *Clostridium acetobutylicum*, which ferments **starch** (maize) to acetone and butanol. Acetone was urgently needed during **World War I** for making **cordite** (smokeless powder). After the war butanol became the main product for lacquers in the car industry.

Organisms

- ***Clostridium acetobutylicum***: Gram-positive, **strict anaerobe**, spore-forming, amylolytic (uses starch directly).
- ***C. beijerinckii*** and *C. saccharobutylicum*, *C. saccharoperbutylacetonicum*: use molasses and sugars; some strains of ***C. beijerinckii*** (formerly *C. butylicum*) produce **isopropanol** instead of acetone (**IBE fermentation**).

Biochemistry: two phases

Phase	Features
Acidogenic phase (growth phase)	Cells grow fast; sugar → pyruvate → acetyl-CoA / butyryl-CoA → **acetic acid and butyric acid**; H_2 and CO_2 released; **pH falls** to about 4.5
Solventogenic phase	At low pH and high acid levels, cells **re-assimilate the acids** and convert them (with more sugar) to **acetone, butanol and ethanol**; pH rises slightly; cells start to **sporulate**. Acetoacetate is decarboxylated to acetone; in IBE strains acetone is reduced to isopropanol by an alcohol dehydrogenase

Typical solvent ratio **acetone : butanol : ethanol = 3 : 6 : 1**. Total solvents only about **2% (about 20 g/L)** because **butanol is toxic** to the cells (damages the membrane at about 1.5–2%); this low titre is the main problem of the process.

Production process

- **Substrate:** maize mash (about 8–10% starch) or molasses (about 6% sugar with added nitrogen); now lignocellulosic hydrolysates.
- **Inoculum:** spores are **heat-shocked** (80–100°C for 1–2 min) to activate germination and kill vegetative contaminants, then grown up anaerobically.
- **Fermentation: anaerobic** (vessel flushed with CO_2 or N_2; the gases formed keep it anaerobic), 33–37°C, initial pH about 6.0–6.5, **no agitation** needed (gas bubbles mix), 36–72 h, batch in large tanks.
- **Gases:** about 60% CO_2 and 40% H_2 by volume are released; they were collected and used.
- **Contamination:** bacteriophages and lactic acid bacteria were major problems; strict sterility is needed.
- **Recovery:** fractional **distillation** of the broth to separate acetone (b.p. 56°C), ethanol (78°C) and butanol (118°C). Modern: in-situ removal by **gas stripping**, pervaporation or liquid–liquid extraction to reduce butanol toxicity.
- **By-products:** spent cells (feed, riboflavin was once recovered), H_2, CO_2.

Isopropyl alcohol (isopropanol)

- Produced by **isopropanol-producing strains of *C. beijerinckii*** (*C. butylicum*) in the IBE fermentation (isopropanol–butanol–ethanol): acetone is further reduced to isopropanol by a secondary alcohol dehydrogenase.
- Engineered *E. coli* expressing clostridial genes and the alcohol dehydrogenase can make isopropanol aerobically at high titres.
- Uses: disinfectant (70% isopropanol), solvent, antifreeze, making acetone and other chemicals. Industrially most isopropanol is still made from propylene.

Feature	ABE fermentation
Organism	*Clostridium acetobutylicum* (starch) / *C. beijerinckii* (sugars; IBE strains)
Oxygen	Strict anaerobe
Temperature, pH	33–37°C; pH starts 6–6.5, falls to about 4.5 in acid phase
Products	Acetone : butanol : ethanol about 3 : 6 : 1; H_2 + CO_2
Main problem	Butanol toxicity → only about 2% total solvents; phage infection; cost of distillation
Recovery	Distillation; gas stripping, pervaporation

Exam point: Mention Weizmann and WWI cordite, the two phases (acidogenic then solventogenic), the 3:6:1 ratio, heat shocking of spores, anaerobic unstirred fermentation and butanol toxicity.

9. INDUSTRIAL PRODUCTION OF VINEGAR

Introduction

Vinegar is a condiment and preservative made by **two successive fermentations** of a sugary or starchy raw material: first an **alcoholic fermentation** by yeast, then an **acetous fermentation** in which acetic acid bacteria oxidize ethanol to **acetic acid**. By law vinegar must contain at least **4 g acetic acid per 100 mL** (4%). The word comes from French *vin aigre* ("sour wine"). Types: wine, cider (apple), malt, spirit (white, from distilled alcohol), rice, cane and fruit vinegars.

Microorganisms

- **Stage 1 (alcohol):** *Saccharomyces cerevisiae* var. *ellipsoideus* converts sugar to about 10–13% ethanol.
- **Stage 2 (acetic acid): acetic acid bacteria**: ***Acetobacter aceti***, *A. pasteurianus*, *Komagataeibacter* (*Gluconacetobacter*) *europaeus* (tolerant to high acid, used in submerged processes), *Gluconobacter oxydans*. They are Gram-negative, **strictly aerobic** rods.
- *Acetobacter* can **over-oxidize** acetic acid to CO_2 and water when ethanol runs out; *Gluconobacter* cannot (lacks a complete TCA cycle).

Biochemistry

Ethanol is oxidized in two steps by membrane-bound enzymes linked to the respiratory chain: ethanol → (**alcohol dehydrogenase**, PQQ-dependent) → acetaldehyde → (**aldehyde dehydrogenase**) → acetic acid.

$C_2H_5OH + O_2 \rightarrow CH_3COOH + H_2O$ (1 g ethanol gives about 1.3 g acetic acid). The reaction needs a **continuous and generous oxygen supply**; even a short interruption of aeration can kill the bacteria in submerged processes. It is strongly exothermic, so cooling is needed.

Methods of production

(1) Orleans (slow, surface) method

- The oldest method (France). Wooden barrels are partly filled with wine plus some old vinegar (as inoculum and to lower pH), leaving air space.
- Acetic acid bacteria form a gelatinous film on the surface called **"mother of vinegar"** (cellulose-producing *Acetobacter xylinum* / *Komagataeibacter xylinus* among others).
- Takes **weeks to months**; part of the vinegar is drawn off and fresh wine added. Gives very good flavour (gourmet wine vinegars) but is slow and needs much space.

(2) Trickling generator (quick vinegar / Schützenbach / Frings generator)

- A tall wooden or steel tank packed with **beech-wood shavings**, corn cobs or coke, on which acetic acid bacteria grow as a film. The tank has a false bottom, air inlets near the base and a sprinkler (sparger) at the top.
- The alcoholic liquid (with some vinegar) is **sprayed at the top** and **trickles down** over the shavings; air enters at the bottom and rises **counter-current**. Bacteria on the shavings oxidize the ethanol.
- The liquid collected at the bottom is **recirculated** (pumped back to the top) until the ethanol is used up (3–5 days). Temperature kept at **about 25–30°C** by cooling coils.
- Advantages: fast, compact, continuous operation. Disadvantages: channelling, slime build-up, and loss of some alcohol by evaporation; lower efficiency than submerged.

(3) Submerged method (Frings Acetator)

- Now the main industrial method (1950s, Hromatka and Ebner, Frings company). A stainless steel fermenter with a special **self-aspirating turbine aerator** (rotor–stator) that draws in air and disperses it as very fine bubbles through the whole liquid; cooling coils; **foam breaker**.
- Bacteria grow **suspended** in the liquid. It runs as a **semi-continuous** process: when ethanol falls to about 0.1–0.2%, about 40–50% of the vinegar is drawn off and replaced with fresh alcoholic mash; each cycle takes about **24–35 h**.

- Conditions: about 30°C, very high aeration; acid up to 12–15% (two-stage processes reach about 20% acetic acid).
- Advantages: **fastest** (about 90–98% efficiency), small space, easy control, high acid strength. Disadvantage: bacteria are very sensitive to **interruption of aeration** (even a few seconds to minutes can kill them), and power cost.

Feature	Orleans (slow)	Trickling generator	Submerged (Acetator)
Bacteria	Surface film	Film on shavings	Suspended in liquid
Aeration	Surface diffusion	Air counter-current	Forced, fine bubbles
Time	Weeks–months	3–5 days	1–2 days per cycle
Efficiency	Low	About 85–90%	About 90–98%
Use	Gourmet wine vinegar	Older industrial	Main modern industrial

Finishing

- **Clarification** by filtration (with bentonite or filter aids) and fining.
- **Ageing** for flavour (esters form).
- **Pasteurization** (60–65°C for about 30 min) or sterile filtration to kill bacteria and prevent "**vinegar eels**" (the nematode *Turbatrix aceti*) and slime.
- Dilution to the desired strength (about 4–6% for table vinegar); bottling.

Uses

- Condiment, pickling and food preservation (the acid lowers pH and inhibits spoilage organisms), salad dressings, sauces and ketchup.
- Manufacture of food-grade acetic acid; cleaning; medicinal and household uses.

Exam point: Two fermentations (yeast then Acetobacter), equation with O_2, the three methods with a diagram of the trickling generator (packing, sprinkler at top, air in at bottom, recirculation) and the table comparing them.

10. INDUSTRIAL PRODUCTION OF ORGANIC ACIDS: LACTIC, CITRIC AND GLUCONIC ACID

Organic acids are among the largest-volume fermentation products. They are **primary metabolites** (or partly oxidized products) made by bacteria and fungi and are used mainly in the food, beverage, pharmaceutical and chemical industries.

A. LACTIC ACID

Introduction and uses

- **Lactic acid (2-hydroxypropanoic acid, $CH_3CHOHCOOH$)** was discovered by Scheele (1780) in sour milk; the first acid produced industrially by fermentation (1881, USA). It exists as **L(+)** and **D(−)** forms; fermentation can give pure L or D, while chemical synthesis gives a racemic (DL) mixture.
- **Uses:** food acidulant and preservative (confectionery, beverages, pickles), **calcium lactate** (calcium supplement), ethyl lactate (solvent), leather tanning, textiles, pharmaceuticals (Ringer lactate), and above all **polylactic acid (PLA)**, a **biodegradable plastic** made from L-lactic acid (the main reason for the recent growth).

Microorganisms

- **Homofermentative lactic acid bacteria** (give only lactic acid, about 90%+ yield through EMP pathway): ***Lactobacillus delbrueckii*** (thermophilic, 45–50°C; subsp. *delbrueckii* uses glucose/sucrose, subsp. *bulgaricus* uses lactose), *L. leichmannii*, *L. casei*, *L. rhamnosus*, *L. amylophilus* (starch), *Lactococcus lactis*, *Streptococcus*, *Bacillus coagulans* (thermophilic, 50–55°C, less contamination).
- Heterofermentative LAB (*Leuconostoc*) give lactic acid + ethanol + CO_2, so they are not used for production.
- Fungus ***Rhizopus oryzae***: aerobic, gives pure L(+)-lactic acid from glucose or starch, easier recovery, but lower yield.

Homolactic fermentation: $\mathbf{C_6H_{12}O_6 \rightarrow 2\ CH_3CHOHCOOH}$ (pyruvate reduced by lactate dehydrogenase; theoretical yield 1 g/g).

Production process

Stage	Details
Substrate	Glucose, sucrose, **molasses**, **whey** (lactose, with *L. bulgaricus*), hydrolysed starch (maize, cassava); 10–15% sugar
Nutrients	LAB are fastidious: need B-vitamins and amino acids, supplied by **corn steep liquor, yeast extract**, malt sprouts
Neutralizer	**$CaCO_3$** (or $Ca(OH)_2$, NH_4OH, NaOH) added to keep **pH 5.5–6.5**, since free lactic acid inhibits the bacteria; lactic acid accumulates as **calcium lactate**
Conditions	**Anaerobic / microaerophilic**, little agitation (only to keep $CaCO_3$ suspended); **45–50°C** for *L. delbrueckii* (high temperature reduces contamination); batch 4–6 days; yield about 90–95% of sugar
Recovery (classical)	Heat broth to 80°C (kill cells, coagulate protein) and raise pH to 10 → filter → concentrate → crystallize **calcium lactate** → add **H_2SO_4** to form lactic acid and insoluble **calcium sulphate (gypsum)**, which is filtered off → decolourize (charcoal) → concentrate. High purity: esterify with methanol, distil methyl lactate and hydrolyse; or ion exchange / electrodialysis
Grades	Technical (22–44%), food (50–80%), pharmaceutical / USP (85–90%), plastic grade (very high purity L-form)

Problem: large amounts of **gypsum** waste (about 1 t per t lactic acid); modern processes use electrodialysis, membranes and low-pH-tolerant strains to avoid it.

B. CITRIC ACID

Introduction and uses

- **Citric acid (2-hydroxy-propane-1,2,3-tricarboxylic acid)** is the most important organic acid made by fermentation (over 2 million t/year). Formerly extracted from lemons (Italy).

- **History:** Wehmer (1893) found *Penicillium* ("*Citromyces*") makes citric acid; **James Currie (1917)** showed ***Aspergillus niger*** produces it in large amounts at low pH; **Pfizer began production in 1923** by surface culture.
- **Uses:** about 70% in **food and beverages** (acidulant in soft drinks, jams, sweets; antioxidant synergist; flavour enhancer), about 12% in **pharmaceuticals** (effervescent tablets, citrate anticoagulant in blood storage, iron citrate), detergents (replacing phosphates), cosmetics, metal cleaning.

Microorganism and biochemistry

- ***Aspergillus niger*** (selected, improved strains) is almost the only commercial organism; yeasts *Yarrowia (Candida) lipolytica* are used on alkanes/oils.
- Citric acid is a **TCA cycle intermediate**: oxaloacetate + acetyl-CoA → citrate (citrate synthase). It **accumulates** when **aconitase** and isocitrate dehydrogenase activity is reduced and the anaplerotic pyruvate carboxylase supplies oxaloacetate.
- **Why these conditions work:** (1) **very low Mn^{2+}** (below 10^{-7} M) changes cell morphology and metabolism, causing overflow; (2) **low Fe** reduces aconitase (an iron-containing enzyme); (3) **high sugar** and high glycolytic flux (feedback inhibition of phosphofructokinase by citrate is relieved by NH_4^+); (4) **low pH (below 2)** suppresses gluconic and oxalic acid formation and contamination; (5) **high aeration**.

Raw materials and medium

- **Beet or cane molasses** (cheapest; must be treated to remove trace metals by **potassium ferrocyanide** [hexacyanoferrate] precipitation or cation exchange), sucrose, glucose syrups, starch hydrolysate.
- High sugar **14–20%**; nitrogen as ammonium salts (limiting, to stop growth); phosphate limited; trace metals controlled (Zn, Mn, Fe low; Cu added to counteract Fe). Methanol (1–3%) is sometimes added to raise yield on crude substrates.

Methods of production

Method	Details
Surface (tray) fermentation	Shallow aluminium or stainless-steel trays (5–20 cm medium) in sterile chambers; spores sprinkled on the surface; mycelial mat forms; 28–30°C, pH 2–3 falling to below 2, 7–15 days; air blown over trays for cooling and O_2. Simple, low energy, but needs large space and labour
Submerged fermentation (main today)	Stirred tank or **airlift** fermenters (up to 500 m^3) of stainless steel (acid resistant); pellet morphology preferred; 28–32°C, **pH below 2** after initial phase, high aeration; 5–10 days; yield about 70–90% of sugar
Solid state (koji) process	Japan: *A. niger* grown on moist wheat bran or sweet-potato residue at 28°C for 5–8 days; citric acid extracted with water

Recovery

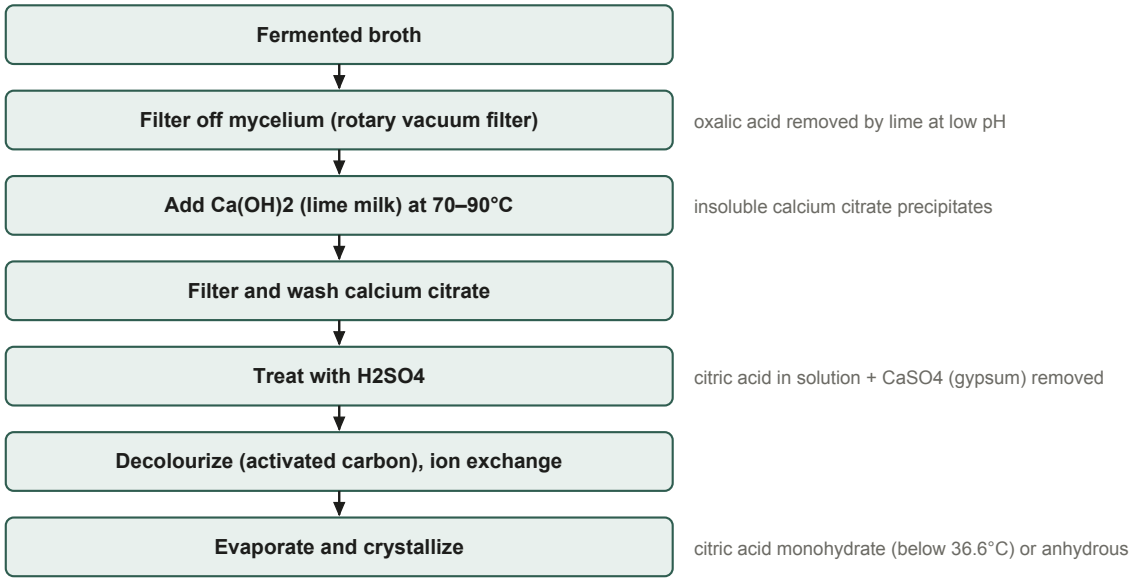


Fig. 6. Recovery of citric acid by the calcium citrate method

Modern plants also use **solvent extraction** (e.g. with tertiary amines) to avoid gypsum.

C. GLUCONIC ACID

Introduction and uses

- **Gluconic acid (D-gluconic acid)** is formed by **oxidation of the aldehyde group (C-1) of glucose** to a carboxyl group. It is mild, non-toxic and non-corrosive, and is an excellent **chelating agent** for Ca, Fe and other metals, especially in alkaline solutions.
- **Uses:** as **sodium gluconate**: sequestering agent in bottle washing, metal cleaning, textile dyeing, concrete (set retarder); **calcium gluconate**: calcium therapy (injections, tablets) and animal feed; **ferrous gluconate**: iron supplement (anaemia); **glucono-δ-lactone (GDL)**: slow acidulant in baking powder, tofu and sausages; zinc and magnesium gluconates in medicine.

Microorganisms and biochemistry

- ***Aspergillus niger*** (main industrial organism): extracellular **glucose oxidase** (FAD enzyme) oxidizes glucose to **glucono-δ-lactone** + H_2O_2; **catalase** breaks down the H_2O_2; the lactone is hydrolysed (by lactonase or spontaneously) to gluconic acid.
- ***Gluconobacter oxydans*** (acetic acid bacterium) oxidizes glucose with a membrane-bound, PQQ-dependent glucose dehydrogenase; also *Penicillium*, *Pseudomonas*, *Zymomonas*.

Glucose + O_2 → glucono-δ-lactone + H_2O_2 → (+H_2O) gluconic acid; 2 H_2O_2 → 2 H_2O + O_2 (catalase). Oxygen is a reactant, so **aeration is critical**.

Production (A. niger, sodium gluconate process)

Point	Details
Medium	High glucose (**about 25–30%**, fed in stages), low nitrogen and phosphate (growth limited, oxidation maximized), corn steep liquor, $MgSO_4$
pH control	Kept at **pH 6.0–6.5** by continuous addition of **NaOH** (gives sodium gluconate) or **$CaCO_3$** (gives calcium gluconate, but Ca gluconate is poorly soluble, so glucose is limited to about 15%). At pH below 3 *A. niger* makes citric acid instead
Conditions	Submerged stirred tank, **very high aeration** (often with raised pressure to increase dissolved O_2), about 30–34°C; very fast: **about 20–40 h**; yield over 95%; mycelium can be reused (semi-continuous)
Recovery	Filter mycelium → decolourize with carbon → concentrate → crystallize **sodium gluconate** (or spray-dry). Calcium gluconate crystallizes on cooling. Free acid by ion exchange or H_2SO_4 treatment of Ca gluconate; GDL by crystallizing from concentrated acid

Feature	Lactic acid	Citric acid	Gluconic acid
Organism	*Lactobacillus delbrueckii*, *Rhizopus oryzae*	*Aspergillus niger*	*A. niger*, *Gluconobacter oxydans*
Oxygen	Anaerobic / microaerophilic	Highly aerobic	Highly aerobic (O_2 is a substrate)
Temperature	45–50°C	28–32°C	30–34°C
pH	5.5–6.5 ($CaCO_3$)	Below 2	6.0–6.5 (NaOH / $CaCO_3$)
Special condition	Neutralization; B-vitamins	Low Mn, low Fe, high sugar	High glucose, very high O_2
Time	4–6 days	5–10 days	20–40 h
Recovery	Calcium lactate + H_2SO_4	Calcium citrate + H_2SO_4	Sodium / calcium gluconate crystallization

Exam point: For each acid: uses → organism → biochemistry (equation / key enzyme) → raw material → conditions (pH, temperature, O_2) → method → recovery. Citric acid: the reason for low Mn and Fe and pH below 2 is a favourite question.

11. VITAMIN PRODUCTION

Vitamins are organic compounds needed in small amounts that the body cannot make. Some are made chemically; **riboflavin** and **vitamin B_{12}** are made mainly by **microbial fermentation** (B_{12} is made **only** by microbes).

Riboflavin (vitamin B_2)

- **Structure and role:** yellow, water-soluble vitamin (isoalloxazine ring + ribitol); forms the coenzymes **FMN** and **FAD** used in redox reactions. Deficiency: ariboflavinosis (cheilosis, glossitis).
- **Producers:** the ascomycete fungi ***Eremothecium ashbyii*** and ***Ashbya gossypii*** (natural overproducers, first used in the 1940s); the yeast *Candida famata* (flareri); and today mostly **genetically engineered *Bacillus subtilis*** and improved *A. gossypii*. Earlier, *Clostridium acetobutylicum* gave riboflavin as a by-product of acetone–butanol fermentation.
- **Biosynthesis:** from **GTP** and **ribulose-5-phosphate**, via 6,7-dimethyl-8-ribityllumazine; glycine and purine precursors increase yield.

Stage	Details (*Ashbya gossypii* process)
Inoculum	Stock → shake flasks → seed tank (about 5%)
Medium	Glucose or sucrose, or **plant oils** (soybean oil); **corn steep liquor**, peptone, collagen or gelatin, yeast extract; glycine (precursor); inositol; no iron limitation needed (unlike *Candida*, which needs **low iron**)
Conditions	Aerobic submerged, stirred tank, about **26–28°C**, pH 6–7, high aeration, about **5–7 days** (fed-batch)
Phases	Growth phase first; riboflavin made mostly after growth slows (**stationary phase**); much of it **crystallizes** in the cells and broth as yellow crystals
Recovery	Heat broth (about 1 h at 120°C) to release riboflavin from cells → centrifuge → collect crystals / precipitate → purify by recrystallization → dry. For feed grade, whole broth is dried
Yield	Modern processes give over 10–20 g/L
Uses	Animal feed supplement (most), human vitamin tablets, **yellow food colour** (E101)

Cyanocobalamin (vitamin B_{12})

- **Structure:** the largest and most complex vitamin: a **corrin ring** with a central **cobalt** atom, a lower ligand **5,6-dimethylbenzimidazole (DMB)** and an upper ligand (cyano-, methyl-, adenosyl- or hydroxo-). Commercial form is **cyanocobalamin**.
- **Role:** coenzyme in methionine synthase and methylmalonyl-CoA mutase; needed for red blood cell formation and nerves. Deficiency: **pernicious anaemia**.
- Made **only by bacteria and archaea**, not by plants or animals, so it is produced entirely by fermentation. First obtained as a by-product of streptomycin fermentation.
- **Producers:** ***Propionibacterium freudenreichii*** (and *P. shermanii*) and ***Pseudomonas denitrificans*** (main industrial organisms, highly improved strains); also *Bacillus megaterium*, *Streptomyces*.

Point	*Propionibacterium freudenreichii* process	*Pseudomonas denitrificans* process
Medium	Glucose, **corn steep liquor**, **cobalt** salt ($CoCl_2$), **DMB** added later	**Sugar beet molasses** (contains **betaine**, which strongly increases yield), CSL, cobalt, DMB
Conditions	**Two-stage**, about 30°C, pH about 7 (neutralized with NH_4OH)	Aerobic throughout, about 30°C, pH 6–7
Stage 1	**Anaerobic** for 2–4 days: growth and formation of the corrin precursor (cobinamide)	Single aerobic stage, fed-batch
Stage 2	**Aerobic** for 3–4 days: DMB is made (or added) and linked, forming complete B_{12}	—
Duration	About 6–7 days	About 5–7 days

- **Recovery:** B_{12} is **intracellular** → cells harvested by centrifugation → heated (80–120°C) at pH 6.5–8.5 to release the vitamin → **potassium cyanide** added to convert all cobalamins to stable **cyanocobalamin** → adsorption/solvent extraction and chromatography → crystallization. Feed grade: concentrated broth dried.
- **Uses:** treatment of **pernicious anaemia** and B_{12} deficiency, vitamin supplements (vegetarians), animal feed.

Other vitamins (extra marks)

- **Vitamin C (ascorbic acid)**: **Reichstein process**: D-glucose → (chemical hydrogenation) D-sorbitol → (***Gluconobacter oxydans*** oxidation) L-sorbose → chemical steps → ascorbic acid. The newer two-step fermentation uses *Gluconobacter* and *Ketogulonicigenium vulgare* to make 2-keto-L-gulonic acid.
- **β-Carotene (provitamin A)**: fungus *Blakeslea trispora* (mixed + and − mating strains), microalga *Dunaliella salina*.
- **Vitamin D**: ergosterol from yeast irradiated with UV gives vitamin D_2.

Exam point: Riboflavin: organisms (Ashbya, Eremothecium, B. subtilis) → medium → conditions → crystallization → recovery → uses. B_{12}: structure with cobalt and DMB → organisms → two-stage anaerobic–aerobic Propionibacterium process → betaine in Pseudomonas → cyanide in recovery → pernicious anaemia.

12. ANTIBIOTIC PRODUCTION

Introduction

Antibiotics are low-molecular-weight **secondary metabolites** produced by microorganisms that, at low concentrations, inhibit the growth of or kill other microorganisms (Waksman). About 70% of known antibiotics come from **actinomycetes** (mainly *Streptomyces*), about 20% from fungi and 10% from other bacteria (*Bacillus*).

Basis	Classes and examples
Producer	Fungi: penicillin, cephalosporin, griseofulvin; Actinomycetes: streptomycin, tetracycline, erythromycin, chloramphenicol; Bacteria: bacitracin, polymyxin
Spectrum	Narrow (penicillin G: Gram-positive), broad (tetracycline, chloramphenicol), antifungal (griseofulvin, nystatin)
Mode of action	Cell wall synthesis (penicillin), protein synthesis (streptomycin, tetracycline), nucleic acid synthesis (rifampicin), cell membrane (polymyxin), mitosis in fungi (griseofulvin)
Chemical structure	β-lactams (penicillins, cephalosporins), aminoglycosides (streptomycin), tetracyclines, macrolides (erythromycin), polyenes (nystatin)

General steps in antibiotic production

1. Strain selection and preservation (high-yield mutant).
2. Inoculum development: spores → shake flask → seed tanks.
3. Production in a sterile, aerated, agitated fermenter, usually **fed-batch**, with trophophase (growth) followed by idiophase (production).
4. Recovery: separation of mycelium, extraction (solvent or ion exchange), purification, crystallization.
5. Quality control: potency (bioassay), sterility, pyrogen and toxicity tests; formulation.

Penicillin

Introduction

- β-lactam antibiotic discovered by **Fleming (1928)**. Basic nucleus is **6-aminopenicillanic acid (6-APA)**: a thiazolidine ring fused with a **β-lactam ring**, with a variable side chain (R-CO-).
- Natural penicillins depend on the precursor in the medium: **penicillin G** (benzylpenicillin, side chain from **phenylacetic acid**) and **penicillin V** (phenoxymethylpenicillin, from **phenoxyacetic acid**, acid-stable, oral).
- Action: inhibits **peptidoglycan cross-linking** (binds transpeptidases / penicillin-binding proteins) in growing bacteria; mainly active on Gram-positive bacteria.

Biosynthesis (short)

L-α-aminoadipic acid + L-cysteine + L-valine → (ACV synthetase) → **δ-(L-α-aminoadipyl)-L-cysteinyl-D-valine (ACV tripeptide)** → (isopenicillin N synthase) → **isopenicillin N** → (acyltransferase exchanges side chain with **phenylacetyl-CoA**) → **penicillin G**.

Production process

Stage	Details
Organism	High-yielding mutants of *Penicillium chrysogenum* (descendants of NRRL 1951 / Wis Q-176)
Inoculum	Spores from lyophilized stock → spore suspension on rice or bran → shake flasks → 1–2 seed fermenters; about 5–10% inoculum, **pellet** growth preferred
Production medium	**Corn steep liquor** (N, growth factors, some phenylacetic acid), **lactose** (slowly hydrolysed, so avoids glucose catabolite repression) or glucose / sucrose fed slowly, ammonium salts, **$CaCO_3$** (buffer), KH_2PO_4, $MgSO_4$, **precursor phenylacetic acid** (fed gradually, as it is toxic above about 0.1%), vegetable oil / antifoam
Fermenter	Stirred tank, 40,000–200,000 L, stainless steel; **fed-batch**; sterile air
Conditions	Temperature **25–27°C**; pH **6.5** (kept 6.4–6.8); high aeration (0.5–1.0 vvm) and agitation; 5–7 days (up to 8)

Stage	Details
Phase 1: trophophase	First 1–2 days: rapid mycelial growth, lactose/glucose used, pH first falls then rises as NH_3 is released; little penicillin
Phase 2: idiophase	Growth slows; **penicillin formed**; continuous feeding of sugar, ammonia and **precursor**; pH kept near 6.5 with acid/alkali
Phase 3	Nutrients exhausted, autolysis of mycelium, pH rises, penicillin production stops; harvest before this

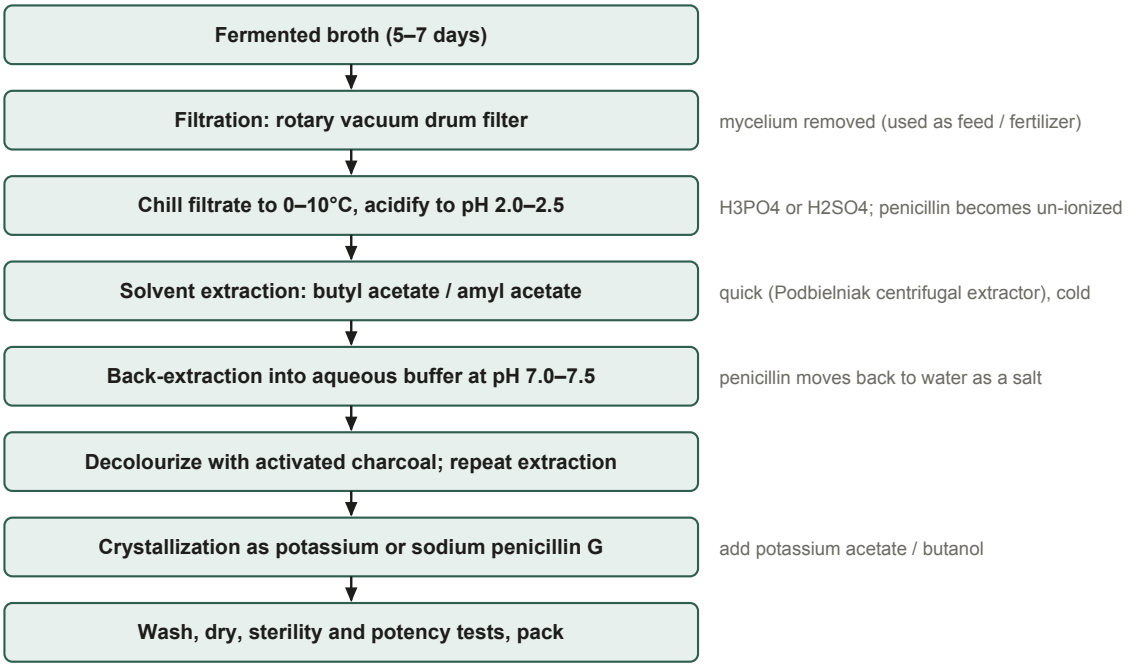


Fig. 7. Recovery (downstream processing) of penicillin

Semi-synthetic penicillins

- Penicillin G is split by **penicillin acylase** (immobilized *E. coli* enzyme) to give **6-APA** and phenylacetic acid.
- New side chains are joined chemically to 6-APA to make **ampicillin, amoxicillin, methicillin, cloxacillin**: broader spectrum, acid stability, or resistance to β-lactamase.

Streptomycin

- **Discovery:** Waksman, Schatz and Bugie, **1944**, from *Streptomyces griseus*.
- **Structure: aminoglycoside** made of three units: **streptidine** (from glucose via myo-inositol), **streptose** and **N-methyl-L-glucosamine**.
- **Action:** binds the **30S ribosomal subunit**, causes misreading of mRNA; active against Gram-negative bacteria and *Mycobacterium tuberculosis*.
- **Medium:** glucose, **soybean meal** (N source), NaCl, (CSL, distillers' solubles); a strain of *S. griseus* as spores → vegetative inoculum.
- **Conditions:** aerobic submerged, stirred tank, **about 28°C**, pH 7.6–8.0, high aeration, **about 7–10 days**.

Phase	What happens
Phase 1 (about 1 day)	Rapid growth of mycelium; proteolysis of soybean meal releases **NH_3**, so pH rises to about 8.0; glucose used slowly; little streptomycin
Phase 2 (days 2–6)	Little growth; **streptomycin accumulates**; glucose and NH_3 used; pH stays 7.6–8.0
Phase 3	Sugar exhausted; mycelium lyses; production stops; pH rises. Harvest at end of phase 2

- **Recovery:** acidify and filter off mycelium → adsorb streptomycin on **activated charcoal** or a **cation-exchange resin** → elute with dilute acid → concentrate → precipitate / crystallize as **streptomycin sulphate** (or calcium chloride complex) → dry.
- **By-product:** vitamin B_{12} can be recovered from the mycelium.

- **Uses:** tuberculosis (with other drugs), plague, tularaemia; in agriculture against plant bacterial diseases.

Tetracyclines

- **Chlortetracycline** (aureomycin, 1948, Duggar) from *Streptomyces aureofaciens*; **oxytetracycline** (terramycin) from *S. rimosus*; **tetracycline** from mutants of *S. aureofaciens* (or by removing Cl from chlortetracycline).
- **Structure:** four fused rings (naphthacene); a **polyketide** made from acetate/malonate units.
- **Action:** bind the **30S subunit** and block attachment of aminoacyl-tRNA; **broad spectrum** (Gram-positive, Gram-negative, rickettsiae, chlamydiae, mycoplasma).
- **Medium:** starch or sucrose, CSL, soybean meal, $CaCO_3$, $(NH_4)_2SO_4$; **phosphate kept low** (it represses production).
- **Key control:** to get **tetracycline** instead of chlortetracycline, **chloride is kept low** in the medium or a **chlorination inhibitor** (bromide, mercaptobenzothiazole) is added, or a chlorination-deficient mutant is used.
- **Conditions:** aerobic submerged, about 28°C, pH about 6.0–6.8, 4–6 days, fed-batch.
- **Recovery:** acidify broth (tetracycline binds Ca/Mg, so oxalic acid is added to release it), filter, extract with butanol or precipitate as a salt, purify and crystallize (as hydrochloride).

Griseofulvin

- **Antifungal** antibiotic from *Penicillium griseofulvum* (also *P. patulum*, *P. urticae*); discovered 1939 (Oxford, Raistrick).
- **Structure:** chlorine-containing **polyketide** (spiro benzofuran); a **chlorine atom** is part of the molecule.
- **Action:** binds **tubulin** and disrupts the mitotic spindle, stopping fungal cell division; deposited in keratin of skin, hair and nails.
- **Production:** aerobic submerged fed-batch fermentation, about 25°C, pH about 6–7, 7–10 days; medium with CSL, lactose or glucose, and **chloride (KCl)** added because it is incorporated into griseofulvin.
- **Recovery:** product is mostly in the mycelium: filter, extract mycelium with organic solvent (butyl acetate / acetone), concentrate and crystallize.
- **Uses:** oral treatment of **ringworm** (dermatophytes: *Trichophyton*, *Microsporum*, *Epidermophyton*) of skin, hair and nails; not active against bacteria or yeasts.

Antibiotic	Producer	Class	Target	Key production point
Penicillin G	*P. chrysogenum*	β-lactam	Cell wall	Phenylacetic acid precursor, lactose, fed-batch, pH 2–2.5 solvent extraction
Streptomycin	*S. griseus*	Aminoglycoside	30S ribosome	Soybean meal, 3 phases, cation exchange recovery
Tetracycline	*S. aureofaciens*, *S. rimosus*	Tetracycline (polyketide)	30S ribosome	Low chloride and low phosphate
Griseofulvin	*P. griseofulvum*	Polyketide (Cl)	Fungal tubulin	Chloride added; extracted from mycelium

Exam point: Penicillin is the most asked long question: organism → structure → inoculum → medium with precursor → fed-batch conditions → phases → recovery recovery flow chart → semi-synthetic penicillins. Short notes on the other three from the summary table.

13. ENZYME PRODUCTION

Why microbial enzymes

- Microbes grow fast, give **high yields** in days, and can be grown in any season.
- Many enzymes are **extracellular**, so extraction is easy and cheap.
- Large variety of enzymes, including ones stable at high temperature (thermophiles), extreme pH (alkaliphiles) or in detergents.
- Yield can be increased by strain improvement and genetic engineering; cheap media.

Types of microbial enzymes

- **Extracellular** (exoenzymes): secreted into the medium, e.g. amylase, protease, lipase, cellulase, pectinase. Easy to recover.
- **Intracellular** (endoenzymes): stay inside cells, e.g. glucose isomerase, glucose oxidase, asparaginase, penicillin acylase. Cells must be broken.
- **Constitutive**: always made. **Inducible**: made only when an **inducer** (usually the substrate) is present, e.g. amylase by starch, cellulase by cellulose.

Regulation that affects yield

- **Induction**: add the inducer to the medium.
- **Catabolite repression**: glucose represses many enzymes, so use slowly used carbon sources or fed-batch feeding.
- **Feedback (end-product) repression**: product accumulation lowers synthesis; use regulatory mutants.
- Improve yield by mutants that are **constitutive** and **catabolite-derepressed**, and by cloning extra gene copies.

Steps in industrial enzyme production

1. **Selection of organism**: high yield, GRAS, stable, short fermentation, no toxins or antibiotics; e.g. *Bacillus*, *Aspergillus*, *Trichoderma*, *Streptomyces*, yeasts.
2. **Medium**: cheap carbon (starch, molasses, bran), nitrogen (soybean meal, CSL, ammonium salts), minerals, **inducer**; pH and C:N ratio optimized.
3. **Sterilization** of medium and fermenter.
4. **Inoculum**: spores or cells → shake flasks → seed tank (about 5–10% v/v).
5. **Fermentation**: SmF (most bacterial enzymes, 2–7 days) or SSF (many fungal enzymes); controlled temperature, pH, aeration; harvest when activity is highest (usually late log or early stationary).
6. **Downstream processing** (flow chart below): separation, cell disruption (for intracellular), concentration, precipitation, purification.
7. **Formulation and standardization**: add stabilizers (glycerol, sorbitol, NaCl, calcium), preservatives; make liquid, powder or dust-free granules (for detergents); adjust to a fixed activity.

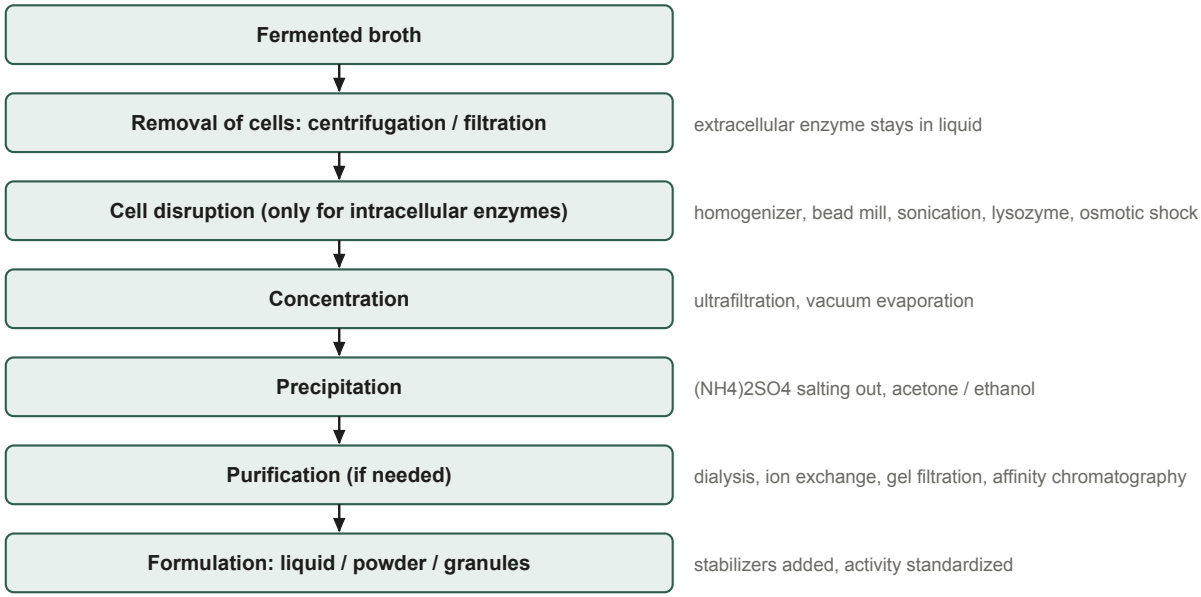


Fig. 8. Downstream processing of microbial enzymes

Examples of enzyme production

α-Amylase (bacterial)

- Organism: *Bacillus amyloliquefaciens*, *B. licheniformis* (thermostable, active at about 90°C), *B. subtilis*.
- Medium: starch (inducer), soybean meal or CSL, $CaCl_2$ (calcium stabilizes amylase); submerged, aerobic, about 30–40°C, pH about 7, 2–4 days.
- Recovery: remove cells, concentrate by ultrafiltration, precipitate or sell as stabilized liquid.
- Fungal amylase is made by *Aspergillus oryzae* in SSF (wheat bran) or SmF.
- Uses: starch liquefaction for glucose syrups, baking, brewing, textile desizing, detergents.

Protease (alkaline, detergent)

- Organism: *Bacillus licheniformis* (subtilisin Carlsberg), *B. subtilis*, alkalophilic *Bacillus*.
- Medium: starch, soybean meal, casein; submerged, aerobic, about 30–37°C, pH 7–9, 2–3 days.
- Recovery: filtration, concentration, precipitation, granulation (dust-free granules to avoid allergy).
- Uses: detergents (largest market), leather dehairing, meat tenderizing, silver recovery from X-ray film.

Enzyme activity and purity

- **Enzyme unit (U)**: the amount of enzyme that converts **1 µmol of substrate per minute** under defined conditions. SI unit: **katal** (1 mol/s).
- **Specific activity** = units per mg protein; rises during purification.
- **Purification fold** = specific activity after step ÷ specific activity of crude extract; **yield (%)** = total units recovered ÷ total units at start × 100.

Important microbial enzymes and applications

Enzyme	Source	Application
α-Amylase	*B. amyloliquefaciens*, *B. licheniformis*, *A. oryzae*	Starch liquefaction, baking, brewing, textile desizing, detergent
Glucoamylase	*Aspergillus niger*, *Rhizopus*	Glucose syrup, brewing, ethanol
Alkaline protease	*B. licheniformis*, *B. subtilis*	Detergents, leather, meat tenderizing
Lipase	*Candida rugosa*, *Rhizopus*, *Aspergillus*, *Pseudomonas*	Detergent, dairy flavour, fat modification, biodiesel
Cellulase	*Trichoderma reesei*, *Aspergillus*	Bioethanol, stone-washed jeans, juice extraction, feed

Enzyme	Source	Application
Pectinase	*Aspergillus niger*	Clarification of fruit juice and wine
Glucose isomerase	*Streptomyces*, *Actinoplanes*	High fructose corn syrup (HFCS)
Invertase	*Saccharomyces cerevisiae*	Invert sugar, soft-centre chocolates
Lactase (β-galactosidase)	*Kluyveromyces lactis*, *Aspergillus*	Lactose-free milk, whey use
Microbial rennet / chymosin	*Rhizomucor miehei*; recombinant in *K. lactis*, *A. niger*	Cheese making
Xylanase	*Trichoderma*, *Aspergillus*	Pulp bleaching, baking, feed
Penicillin acylase	*E. coli*, *Bacillus megaterium*	6-APA for semi-synthetic penicillins

Therapeutic enzymes

Enzyme	Source	Use
L-Asparaginase	*E. coli*, *Erwinia chrysanthemi*	Acute lymphoblastic leukaemia (removes asparagine needed by tumour cells)
Streptokinase	β-haemolytic *Streptococcus*	Dissolves blood clots (heart attack), activates plasminogen
Urokinase	Human kidney cells / recombinant *E. coli*	Thrombolytic (clot dissolving)
Serratiopeptidase	*Serratia marcescens*	Anti-inflammatory
Collagenase	*Clostridium histolyticum*	Wound debridement, burns
Lysozyme	Egg white / microbial	Antibacterial

Analytical enzymes

- **Glucose oxidase** (*A. niger*) + peroxidase: blood and urine glucose tests, glucose **biosensors** (glucometer strips).
- **Cholesterol oxidase** (*Streptomyces*, *Nocardia*): serum cholesterol test. **Urease**: blood urea. **Luciferase**: ATP detection (hygiene tests).
- **Horseradish peroxidase** and **alkaline phosphatase**: enzyme labels in **ELISA**.
- Molecular biology: **restriction endonucleases**, DNA ligase, **Taq polymerase** (*Thermus aquaticus*, PCR), reverse transcriptase.

Exam point: Answer order: why microbes → types (extra/intracellular, inducible) → steps with medium and conditions → downstream flow chart → one example (amylase or protease) → therapeutic / analytical / industrial applications table.

14. IMMOBILIZATION OF ENZYMES

Definition

Enzyme immobilization is the attachment or confinement of an enzyme (or whole cells) on or within an inert, insoluble **support (matrix)** so that it keeps its catalytic activity and can be **used repeatedly or continuously**. The first example was **invertase adsorbed on charcoal** (Nelson and Griffin, **1916**); the first industrial process was **aminoacylase** on DEAE-Sephadex for L-amino acids (**Chibata, Tanabe Seiyaku, Japan, 1969**).

Advantages

- **Reuse** of the enzyme many times, which lowers cost.
- Allows **continuous** processes in packed bed or fluidized bed reactors.
- Product is **not contaminated** by enzyme, so purification is easy.
- Often **greater stability** to temperature, pH, organic solvents and storage.
- Reaction can be **stopped quickly** by removing the enzyme; better process control.
- High enzyme concentration per reactor volume.

Disadvantages

- **Loss of activity** during immobilization (active site blocked or changed).
- **Diffusion (mass transfer) limitation**: substrate must reach the enzyme; big substrates move slowly.
- Extra cost of support and process.
- Leakage of enzyme from some supports; microbial contamination in long runs.
- Not useful when the substrate is insoluble or very large.

Properties of a good support

- Inert, non-toxic, insoluble, cheap and easily available.
- Mechanically strong, chemically and microbially stable.
- Large surface area, porous, allowing substrate and product to diffuse.
- Has groups for binding (-OH, -NH_2, -COOH); can be regenerated.

Type of support	Examples
Natural organic polymers	Alginate, agar, agarose, carrageenan, cellulose, chitosan, gelatin, collagen, starch
Synthetic organic polymers	Polyacrylamide, polystyrene, nylon, polyvinyl alcohol, ion-exchange resins
Inorganic materials	Activated charcoal, porous glass, silica, alumina, clay, bentonite

Methods of immobilization

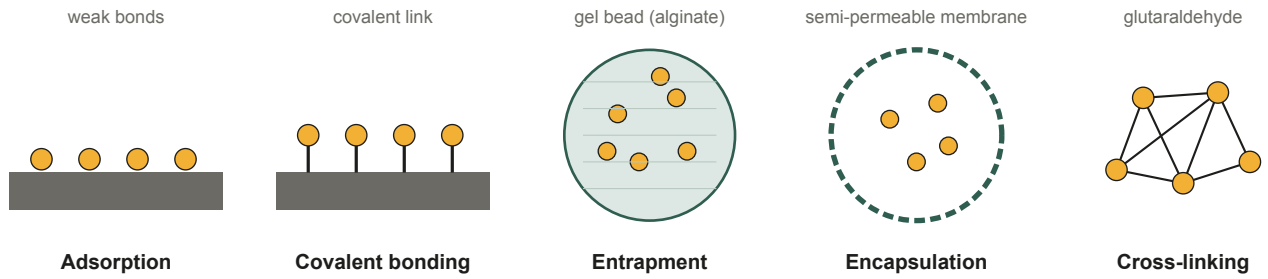


Fig. 9. Methods of enzyme immobilization (enzyme shown as yellow circles)

(1) Adsorption

- Enzyme is bound to the surface of the support by **weak forces**: van der Waals, ionic, hydrogen and hydrophobic bonds.

- **Procedure:** mix enzyme solution with the support under suitable pH and ionic strength, then wash off unbound enzyme.
- **Supports:** activated charcoal, alumina, silica, clay, glass, ion-exchange materials (DEAE-cellulose, CM-cellulose, DEAE-Sephadex).
- **Merits:** simple, cheap, mild, little change in enzyme, support can be reused. **Demerits:** enzyme **leaks (desorbs)** with change in pH, temperature or ionic strength.

(2) Covalent bonding

- Enzyme is joined by **covalent bonds** between its functional groups (-NH_2 of lysine, -COOH, -SH, -OH) and a **chemically activated** support. Groups needed for catalysis must not be used.
- **Activating agents:** cyanogen bromide (CNBr) for agarose/Sepharose and cellulose, **glutaraldehyde**, carbodiimide, silanes for glass.
- **Merits:** strong binding, **no leakage**, stable. **Demerits:** harsh chemistry can **inactivate** the enzyme; support cannot be reused; costly.

(3) Entrapment

- Enzyme or whole cells are **physically trapped** inside the lattice of a polymer gel or fibres; small substrates and products diffuse through the pores, the enzyme cannot escape.
- **Calcium alginate method (most common):** mix enzyme or cells with **2–3% sodium alginate** → drop through a syringe into **0.1–0.2 M $CaCl_2$** → Ca^{2+} cross-links alginate and forms **beads** → harden for 30 min, wash.
- Other gels: **polyacrylamide** (polymerized with bis-acrylamide), κ-carrageenan (with KCl), agar, gelatin, polyvinyl alcohol.
- **Merits:** mild, simple, suits **whole cells**, no chemical change of enzyme. **Demerits:** small enzymes can leak; diffusion limitation; alginate dissolves with phosphate or citrate (chelators).

(4) Encapsulation (microencapsulation)

- Enzyme is enclosed inside **semi-permeable membrane capsules** (microcapsules about 10–100 μm) or liposomes. Only small molecules pass through the membrane.
- **Membranes:** nylon, cellulose nitrate, polyethyleneimine; lipid vesicles (liposomes); hollow fibre membranes.
- **Merits:** large amount of enzyme held; several enzymes can be co-encapsulated. **Demerits:** only for small substrates; capsules may break.

(5) Cross-linking (carrier-free)

- Enzyme molecules are joined **to each other** by covalent bonds using **bifunctional reagents**, forming insoluble aggregates without any support.
- **Reagent: glutaraldehyde** (most common), hexamethylene diisocyanate; modern forms: CLEC (cross-linked enzyme crystals), **CLEA** (cross-linked enzyme aggregates).
- **Merits:** simple, no support cost, high enzyme density. **Demerits:** loss of activity, gel-like aggregates weak and hard to handle; often combined with adsorption.

Feature	Adsorption	Covalent	Entrapment	Encapsulation	Cross-linking
Binding force	Weak physical	Covalent	Physical trapping	Membrane barrier	Covalent (enzyme-enzyme)
Preparation	Simple	Difficult	Moderate	Moderate	Simple
Activity loss	Low	High	Low	Low	Moderate–high
Enzyme leakage	Yes	No	Possible	No (small enzymes may)	No
Cost	Low	High	Low	Moderate	Low

Changes in properties after immobilization

- **Stability** usually increases (thermal, pH, storage).

- **Optimum pH** may shift because of the charge of the support (microenvironment effect).
- **Apparent K_m** often increases because of diffusion limitation; V_{max} may fall.
- Activity towards large substrates falls (steric hindrance).

Immobilized whole cells

- Cells are immobilized (usually in alginate or carrageenan) when the enzyme is intracellular, unstable when purified, or needs cofactors.
- Advantages: no enzyme extraction, multi-step reactions possible, cofactors regenerated. Examples: immobilized *E. coli* (aspartase) for L-aspartic acid (Tanabe, 1973); *Brevibacterium* (fumarase) for L-malic acid; immobilized yeast for ethanol and beer; *Rhodococcus* (nitrile hydratase) for acrylamide.

Applications

Immobilized enzyme / cell	Product / use
Glucose isomerase (*Streptomyces*)	Glucose → fructose: **high fructose corn syrup**; the largest immobilized enzyme process
Aminoacylase (*Aspergillus oryzae*)	Resolution of DL-amino acids to L-amino acids (L-methionine, L-phenylalanine); first industrial process 1969
Penicillin G acylase (*E. coli*)	Penicillin G → **6-APA** for ampicillin, amoxicillin
Lactase (β-galactosidase)	Lactose-free milk; whey hydrolysis
Invertase	Invert sugar syrup
Nitrile hydratase (*Rhodococcus* cells)	Acrylonitrile → **acrylamide**
Aspartase (*E. coli* cells)	L-aspartic acid (for aspartame sweetener)
Glucose oxidase on electrode	**Biosensors** for blood glucose
Urease in microcapsules	Artificial kidney (removal of urea)

Exam point: Draw the immobilization figure and describe each method with principle, support/agent, merits and demerits. Mention the Ca-alginate bead procedure, glutaraldehyde, and glucose isomerase → HFCS as the main application.

15. SINGLE CELL PROTEIN (SCP)

Definition and history

Single cell protein (SCP) is the **dried microbial biomass** (or protein extracted from it) of bacteria, yeasts, filamentous fungi or algae, grown on large scale and used as a **protein source in human food or animal feed**. The term was coined by **Carroll L. Wilson (MIT, 1966)**, because the organisms are single-celled or simple. The idea is old: yeast was eaten as food in Germany during both World Wars, and *Spirulina* has been eaten for centuries by people around Lake Chad and by the Aztecs.

Why SCP (advantages)

- Very **fast growth**: biomass doubles in hours (bacteria 20 min–2 h, yeast 1–3 h) compared with weeks for plants and animals.
- **High protein content** (40–80% of dry weight) with a good amino acid profile (rich in lysine).
- Can be grown on **cheap waste materials** (molasses, whey, straw, sewage, methanol), converting waste to food.
- Needs little land and water; production is independent of climate and season; easy genetic improvement.
- Also rich in **vitamins** (B-complex) and minerals; some contain useful lipids and pigments.

Microorganisms and substrates

Group	Organisms	Substrates	Protein (% dry wt)
Bacteria	*Methylophilus methylotrophus* (ICI Pruteen), *Methylococcus capsulatus*, *Pseudomonas*, *Cellulomonas*, *Rhodopseudomonas*	Methanol, methane (natural gas), n-alkanes, cellulose	60–80
Yeasts	*Candida utilis* (torula yeast), *C. tropicalis*, *C. lipolytica*, *Saccharomyces cerevisiae*, *Kluyveromyces fragilis*	Molasses, **whey**, sulphite waste liquor, n-alkanes, ethanol	45–55
Filamentous fungi	***Fusarium venenatum*** (Quorn mycoprotein), *Chaetomium cellulolyticum*, *Paecilomyces variotii* (Pekilo process), *Trichoderma*, mushrooms	Starch, glucose, sulphite liquor, lignocellulose	30–50
Algae	***Spirulina*** (*Arthrospira platensis*), ***Chlorella***, *Scenedesmus*	CO_2 + sunlight (photosynthetic) in ponds	50–70

Substrates used for SCP production

- **Carbon dioxide** with sunlight (algae in open ponds or photobioreactors).
- **Hydrocarbons:** n-alkanes and gas oil (BP "Toprina" with yeast in the 1970s), methane (natural gas, *Methylococcus*).
- **Methanol:** cheap, pure, water-miscible (ICI Pruteen with *Methylophilus methylotrophus*).
- **Ethanol**: very clean substrate for food-grade yeast.
- **Carbohydrates and wastes:** cane/beet molasses, **whey** from cheese making, starch wastes (potato, cassava), fruit and vegetable wastes, **sulphite waste liquor** from paper mills (Pekilo), **lignocellulose** (straw, bagasse) after pre-treatment, sewage and animal waste (for feed only).

Production process

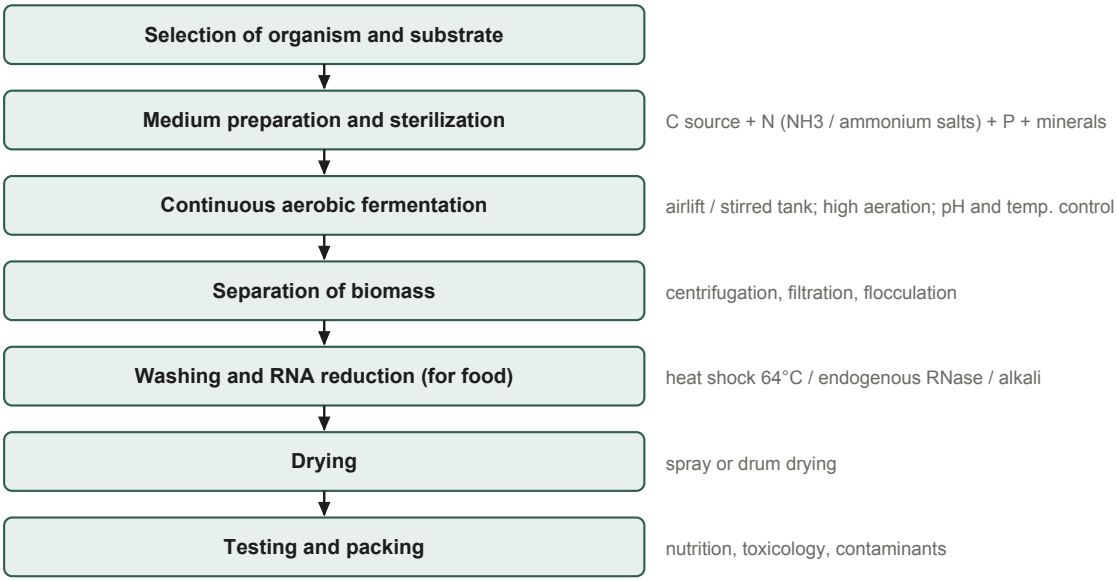


Fig. 10. General process of SCP production

- Most SCP processes are **aerobic continuous** fermentations (chemostat) because they make growth-linked biomass; high oxygen demand and heat removal are the main engineering problems (e.g. the ICI Pruteen **pressure-cycle airlift fermenter** of 1500 m^3, UK, 1980).
- **Quorn mycoprotein** (*Fusarium venenatum*, UK, 1985): continuous airlift fermentation on glucose; RNA reduced by heating to 64°C; the fungal hyphae give a meat-like texture; sold as a meat substitute.
- **Spirulina**: grown in shallow raceway ponds with high pH (about 10) and bicarbonate (few competitors), harvested by filtration, dried; sold as health food tablets.

Nutritional value

Component	Details
Protein	40–80% of dry weight; good essential amino acids, rich in **lysine** (useful with cereals), often low in **sulphur amino acids** (methionine, cysteine)
Vitamins	Rich in B-complex (thiamine, riboflavin, niacin, pyridoxine, B_{12} in some); algae give β-carotene
Lipids	Algae and some yeasts have useful fatty acids (Spirulina: γ-linolenic acid)
Minerals	Good source of P, K, iron
Digestibility	Variable; cell walls (especially algae, yeasts) reduce digestibility unless broken

Problems and limitations

- **High nucleic acid content** (bacteria 8–16%, yeast 6–10%, fungi 2.5–6%, algae 4–6%). RNA purines are metabolized to **uric acid**; humans lack uricase, so excess causes **gout and kidney stones**. Humans should take no more than about **2 g of nucleic acid per day**. RNA is reduced by **heat shock** (activating the cell's own RNases), alkaline hydrolysis or adding RNase.
- Possible **toxins** or carcinogens from substrates (hydrocarbon residues), and mycotoxins from fungi; must pass long toxicological tests.
- Bacteria may contain **endotoxins** and are small, so harvesting is costly; possible allergic reactions, gastrointestinal upset, skin reactions.
- Poor digestibility of cell walls; **low acceptability** (colour, taste, texture) and high production cost compared with soybean and fish meal. These economic reasons closed Pruteen and Toprina in the 1980s.

Uses

- **Animal feed:** poultry, pigs, fish and calf milk replacers (the main use).
- **Human food:** Quorn mycoprotein, Spirulina and Chlorella tablets, food yeast (nutritional yeast, yeast extract), protein supplements for undernourished populations.

- Newer: methane- and CO_2-based bacterial protein for fish feed, "air protein" from hydrogen-oxidizing bacteria.

Exam point: Definition (Carroll Wilson 1966) → advantages → organism and substrate table → process flow chart → nutritional value → problems (nucleic acid 2 g/day, uric acid, toxins, acceptability) → examples Pruteen, Quorn, Spirulina.

16. BIOTRANSFORMATION (MICROBIAL TRANSFORMATION)

Definition

Biotransformation (bioconversion, microbial transformation) is the use of microorganisms, their spores, or their enzymes to bring about a **specific chemical change** in a compound that is added to the culture, converting it into a structurally related, more useful product. Usually only **one or a few steps** are catalysed (a single enzyme reaction), unlike fermentation where the product is made from simple nutrients through many reactions.

Features and advantages over chemical methods

- **High specificity**: **regio-specific** (acts on one particular position of a molecule, e.g. only C-11 of a steroid) and **stereo-specific** (gives only one optical isomer).
- Works under **mild conditions** (room temperature, normal pressure, neutral pH), so less energy and fewer side reactions.
- Can do reactions that are very difficult chemically (e.g. hydroxylation at an unactivated carbon).
- Fewer steps, less toxic waste ("green chemistry"), cheaper for complex molecules.

Limitations

- Low solubility of substrates (steroids) in water; toxicity of substrate or product to cells.
- Side reactions by other enzymes of the organism; product may be further degraded.
- Low concentration and yield; recovery from dilute broth.

Types of reactions

Reaction	Example
Oxidation / hydroxylation	11α-hydroxylation of progesterone by *Rhizopus*; sorbitol → sorbose by *Gluconobacter*; ethanol → acetic acid
Dehydrogenation	Δ^1-dehydrogenation of cortisol → prednisolone by *Arthrobacter simplex*
Reduction	Ketone → chiral alcohol by baker's yeast
Hydrolysis	Penicillin G → 6-APA (penicillin acylase); ester and amide hydrolysis
Isomerization	Glucose → fructose (glucose isomerase)
Condensation / addition	Benzaldehyde + acetaldehyde (from yeast) → L-phenylacetylcarbinol (L-PAC, precursor of **ephedrine**) by *S. cerevisiae*
Amination, decarboxylation, methylation, side-chain cleavage	Fumarate + NH_3 → L-aspartate (aspartase); sterol side-chain cleavage by *Mycobacterium*
Resolution of racemic mixtures	DL-amino acids → L-amino acids (aminoacylase)

Steroid biotransformations (most important)

- **1952: Peterson and Murray** (Upjohn) found that ***Rhizopus arrhizus*** and ***R. nigricans*** hydroxylate **progesterone at C-11α** in one step, giving 11α-hydroxyprogesterone, the key intermediate for **cortisone**. The chemical route needed 31 steps; microbial conversion cut the price of cortisone drastically. This launched industrial biotransformation.
- **11β-hydroxylation**: *Curvularia lunata* converts Reichstein's compound S (11-deoxycortisol) to **hydrocortisone (cortisol)**.
- **Δ^1-dehydrogenation**: ***Arthrobacter simplex*** (*Nocardioides simplex*) and *Septomyxa affinis* convert cortisol to **prednisolone** (more anti-inflammatory, less salt retention).
- **Side-chain cleavage** of cheap plant sterols (sitosterol, cholesterol) by *Mycobacterium* mutants gives **androstenedione (AD / ADD)**, starting material for sex hormones and corticosteroids.
- **Diosgenin** (from Mexican yam) and stigmasterol (soybean) are the plant raw materials for steroid drugs.

Substrate	Organism	Product	Reaction
Progesterone	*Rhizopus arrhizus* / *R. nigricans*	11α-hydroxyprogesterone (→ cortisone)	11α-hydroxylation
11-Deoxycortisol (compound S)	*Curvularia lunata*	Hydrocortisone (cortisol)	11β-hydroxylation
Cortisol	*Arthrobacter simplex*	Prednisolone	Δ^1-dehydrogenation
Cortisone	*A. simplex*, *Septomyxa*	Prednisone	Δ^1-dehydrogenation
Sitosterol	*Mycobacterium* mutants	Androstenedione	Side-chain cleavage

Other examples

- **Vitamin C (Reichstein process):** D-sorbitol → **L-sorbose** by *Gluconobacter oxydans* (regio-specific oxidation at C-5), the only microbial step; the rest is chemical.
- **L-PAC for ephedrine** (a nasal decongestant) by yeast pyruvate decarboxylase condensing benzaldehyde with acetaldehyde.
- **Semi-synthetic penicillins** via 6-APA; **7-ADCA/7-ACA** for cephalosporins.
- **Acrylamide** from acrylonitrile by nitrile hydratase of *Rhodococcus rhodochrous* J1 (Nitto, Japan; over 100,000 t/year).
- L-aspartic acid (aspartase, *E. coli*), L-malic acid (fumarase), 1,3-propanediol from glycerol, antibiotics and alkaloid modification, detoxification of pollutants.

Methods used in biotransformation

Method	Description
Growing culture	Substrate added to an actively growing culture (usually at the end of log phase) dissolved in ethanol/acetone/DMF because steroids are poorly soluble
Resting (washed) cells	Grown cells are harvested and suspended in buffer with the substrate; fewer side products, easy recovery
Spores	Fungal spores (e.g. *Aspergillus*, *Penicillium*) carry enzymes and can transform steroids without growing
Immobilized cells or enzymes	Reused in continuous reactors (e.g. immobilized *A. simplex* for prednisolone)
Purified enzymes	Cleanest, but costly; used when cofactors are not needed
Two-phase / organic solvent systems	Substrate dissolved in an organic phase in contact with cells, to overcome low solubility

General procedure

1. **Screening** of organisms able to carry out the desired reaction (culture collections, soil).
2. Growth of the organism under optimum conditions to produce high enzyme activity (sometimes induced by adding substrate).
3. Addition of the substrate at the right time and concentration (in a water-miscible solvent or as a fine suspension).
4. Incubation with monitoring (TLC, HPLC) until conversion is complete.
5. **Extraction** of product with solvents (chloroform, ethyl acetate for steroids), purification by chromatography and crystallization.

Exam point: Definition → how it differs from fermentation → advantages (regio- and stereo-specificity, mild conditions) → reaction types table → steroid examples with organisms (Peterson and Murray 1952, Rhizopus, Arthrobacter simplex, Curvularia) → sorbose → methods.

17. PRACTICALS (SYLLABUS)

1. Study of fermenter and fermenter operation

- **Aim:** to study the parts of a laboratory fermenter (bench-top, 2–10 L) and its operation.
- **Parts to identify:** glass vessel with stainless-steel head plate, motor and agitator shaft with impellers, sparger, baffles, cooling/heating jacket or coil, air pump and sterile air filter (0.2 µm), exhaust condenser, probes (pH, DO, temperature, foam), peristaltic pumps for acid/base/antifoam/feed, sampling port, control unit (display and set points).
- **Operation:** (1) calibrate pH probe (buffers 4 and 7) before sterilization; (2) prepare and fill medium; (3) sterilize vessel with medium in an autoclave at 121°C, 15 psi, 20–30 min (or in-situ steam); (4) connect air, cooling water and pumps aseptically; (5) set temperature, pH, agitation and aeration; calibrate DO probe (0% with N_2, 100% with air); (6) inoculate aseptically (5–10% v/v); (7) take samples at intervals for OD, dry weight, substrate and product; (8) harvest, clean and dry.
- **Observations:** plot growth curve, pH and DO changes with time; note foam and antifoam use.

2. Isolation and purification of important microbial enzymes

- **Isolation (screening):** plate soil dilutions on specific media: starch agar (amylase, iodine test), skim milk agar (protease), tributyrin agar (lipase), CMC agar with Congo red (cellulase). Measure zone diameter; select best isolates.
- **Production:** grow selected isolate in production broth in shake flasks (e.g. 37°C, 150 rpm, 24–72 h).
- **Crude enzyme:** centrifuge at 8,000–10,000 rpm for 10–15 min at 4°C; supernatant = crude extracellular enzyme.
- **Purification steps:** (a) **ammonium sulphate precipitation** (add salt slowly on ice to 60–80% saturation, keep overnight at 4°C, centrifuge, dissolve pellet in minimum buffer); (b) **dialysis** against buffer to remove salt; (c) optional ion-exchange or gel filtration chromatography; (d) check by SDS-PAGE.
- **Assay:** measure enzyme activity (e.g. DNS method for reducing sugar released by amylase; titration or p-nitrophenyl palmitate for lipase) and protein (**Lowry** or **Bradford**). Prepare a purification table: total activity, total protein, specific activity (U/mg), purification fold, yield %.

3. Production of lipase by submerged fermentation

- **Organism:** lipolytic bacterium or fungus (e.g. *Bacillus*, *Pseudomonas*, *Aspergillus*, *Candida*) screened on tributyrin agar.
- **Medium:** peptone / yeast extract, mineral salts, **olive oil 1% (inducer)** emulsified with Tween 80 or gum arabic; pH about 7.
- **Procedure:** inoculate (2–5% v/v) into 100 mL medium in 250 mL flasks; incubate at 30–37°C on shaker (150 rpm) for 24–96 h; harvest by centrifugation; supernatant is crude lipase.
- **Assay (titrimetric):** olive oil emulsion + buffer + enzyme at 37°C for 30 min; stop with acetone–ethanol; titrate liberated fatty acids with 0.05 N NaOH using phenolphthalein. One unit = 1 µmol fatty acid released per minute. (Alternatively spectrophotometric with p-nitrophenyl palmitate at 410 nm.)

4. Production of lipase by semi-solid (solid state) fermentation

- **Substrate:** agro-residues such as wheat bran, rice bran or oil cakes (groundnut, coconut, mustard) that contain some lipid.
- **Procedure:** 5–10 g substrate in 250 mL flask, moistened with mineral salt solution (with or without 1–2% olive oil) to 50–70% moisture (semi-solid if more water), autoclave, cool, inoculate with spore suspension (fungi) or culture, incubate at 28–30°C for 3–7 days without shaking (mix gently daily).
- **Extraction:** add buffer (1:5–1:10 w/v, e.g. phosphate buffer pH 7 with 0.1% Tween 80), shake 30–60 min, filter through muslin and centrifuge; the extract is crude enzyme. Assay as above and express activity as **U per gram dry substrate (U/gds)**.

- **Compare** yields of SmF and SSF: SSF usually gives a more concentrated enzyme with cheaper substrates.

5. Effect of pH and temperature on enzyme production

- **pH:** prepare production medium at different pH values (e.g. 4, 5, 6, 7, 8, 9) using suitable buffers; inoculate equally; incubate at a fixed temperature; assay enzyme activity; plot **activity vs pH** to find the **optimum pH**.
- **Temperature:** incubate identical flasks at different temperatures (e.g. 20, 25, 30, 37, 45, 55°C) at optimum pH; assay; plot **activity vs temperature** to find the **optimum temperature**.
- **Result:** bell-shaped curves. Activity rises to an optimum and then falls because of denaturation (temperature) or change in ionization of the enzyme and substrate (pH). Record growth (OD) alongside, since optimum for growth and for enzyme production may differ.

18. QUICK REVISION AND QUESTION BANK

Remember	Answer
Term "biotechnology"	Karl Ereky, 1919
Father of industrial microbiology	Louis Pasteur
Cell-free fermentation	Eduard Buchner, 1897
Penicillin / streptomycin	Fleming 1928 / Waksman 1944
Airlift zones	Riser (gassed) and downcomer; draft tube
Chemostat	D = F/V; μ = D at steady state; washout if D > μ_{max}
SSF	Little free water, fungi, bran, koji; water activity
Ethanol equation / yield	$C_6H_{12}O_6 \rightarrow 2\ C_2H_5OH + 2\ CO_2$; 0.511 g/g
Rectified spirit	95.6% ethanol, azeotrope; absolute alcohol by molecular sieve
Melle–Boinot process	Ethanol with yeast recycling
Neuberg second / third form	Sulphite binds acetaldehyde / alkaline dismutation → glycerol
ABE organism and ratio	*Clostridium acetobutylicum*; acetone : butanol : ethanol = 3 : 6 : 1
Isopropanol producer	Isopropanol strains of *C. beijerinckii* (IBE)
Vinegar organisms	Yeast, then *Acetobacter aceti* / *Komagataeibacter*
Vinegar methods	Orleans (slow), trickling generator (quick), Frings Acetator (submerged)
Lactic acid organism / temperature	*Lactobacillus delbrueckii*, 45–50°C, $CaCO_3$, recovered as calcium lactate
Citric acid conditions	*A. niger*, low Mn and Fe, high sugar, pH below 2; ferrocyanide treatment of molasses
Gluconic acid enzyme	Glucose oxidase (+ catalase), *A. niger*, pH 6–6.5, very high O_2
Riboflavin / B_{12}	*Ashbya gossypii* / *Propionibacterium*, *Pseudomonas denitrificans* (cobalt, DMB)
Penicillin precursor / extraction	Phenylacetic acid; pH 2–2.5, butyl acetate
Tetracycline / griseofulvin	Low chloride / chloride added
Enzyme unit	1 μmol per minute; katal = mol/s
Immobilization	Ca-alginate entrapment; glutaraldehyde cross-linking; glucose isomerase → HFCS
SCP term / problem	Carroll Wilson 1966; nucleic acid → uric acid, max 2 g/day
SCP examples	Pruteen (*Methylophilus*, methanol), Quorn (*Fusarium venenatum*), Spirulina
Biotransformation landmark	Peterson and Murray 1952, *Rhizopus*, 11α-hydroxyprogesterone
Prednisolone	Cortisol Δ^1-dehydrogenation by *Arthrobacter simplex*
Sorbose (vitamin C)	Sorbitol oxidation by *Gluconobacter oxydans*

Long answer questions

1 Define biotechnology. Give its history and scope.

2 Describe the design of a stirred tank fermenter. Write notes on airlift, bubble column, fluidized bed, packed bed, hollow fibre and photobioreactors.

3 Differentiate batch, fed-batch and continuous fermentation; compare submerged and solid state fermentation.

4 Describe the industrial production of ethanol from molasses and its recovery.

5 Explain Neuberg's steered fermentations for glycerol production.

6 Describe the acetone–butanol fermentation. Add a note on isopropanol.

7 Describe the methods of vinegar production with a diagram of the trickling generator.

8 Describe the industrial production of citric acid. Why are low manganese, low iron and low pH needed?

9 Write the production and recovery of lactic acid and gluconic acid.

10 Describe the production of riboflavin and vitamin B_{12}.

11 Describe the production and recovery of penicillin. Write notes on streptomycin, tetracycline and griseofulvin.

12 Describe enzyme production and the therapeutic, analytical and industrial uses of microbial enzymes.

13 What is immobilization? Describe the methods with merits, demerits and applications.

14 What is single cell protein? Discuss organisms, substrates, nutritional value and limitations.

15 Define biotransformation. Discuss steroid transformations with examples.

16 Explain screening and strain improvement of industrial microorganisms.

Short notes

- Karl Ereky · Weizmann · Peoria · Buchner
- Baffles · sparger · $k_L a$ · washout · turbidostat · koji
- Fusel oil · azeotrope · Melle–Boinot · spent wash
- Mother of vinegar · Frings Acetator · vinegar eels
- Gypsum problem · ferrocyanide · glucono-δ-lactone
- Heat-shocked spores · butanol toxicity
- Nucleic acid in SCP · Quorn · Spirulina
- L-PAC · 11α-hydroxylation · prednisolone
- CLEA · Ca-alginate · enzyme unit

References: Stanbury, Whitaker and Hall, Principles of Fermentation Technology; Casida, Industrial Microbiology; Patel, Industrial Microbiology; Prescott and Dunn's Industrial Microbiology (Reed); Crueger and Crueger, Biotechnology: A Textbook of Industrial Microbiology; Waites et al., Industrial Microbiology: An Introduction; class notes on history of biotechnology and bioreactor design.